Secondary selection at the critical clock of the literal IGATH-2 recursion

Abstract

For the literal IGATH-2 recursion with fixed gradient step $1/4$, the critical schedule $h_k = k^{-1/3}$ makes the normal-stiffness drift act for infinite harmonic time. Near a compact C^5 minimum manifold of a C^5 objective with normal spectrum in $[\mu, L] \subset (0, 8)$, we prove late-history capture using a collision-safe normal metric and a cumulative-transient velocity barrier. A Morse–Bott reduction of the restricted squared norm then yields an actual center step $\operatorname{grad}_S \psi/(6k) + O(k^{-4/3})$, where $\psi = \langle Np, H^{-1}Np\rangle/2$, together with summable normal-history errors. Secondary values converge and every cluster point is critical without a gradient inequality. An additional one-sided Kurdyka–Łojasiewicz condition on $-\psi|_S$ gives a single point limit and finite length of both actual paths. On a variable-stiffness circle, a separate uniform argument selects the minimum-stiffness point from every compact subarc away from the maximum, with fixed-entry rate $O(k^{-1/54})$ and a finite signed scaled limit. The same prepared history has different limits under the reciprocal and critical schedules. All entries and estimates retain the original global indices.

Keywords: IGATH-2; critical regularization clock; minimum manifold; secondary selection; normal compliance potential; finite path length

2020 Mathematics Subject Classification. Primary 37N40; Secondary 39A30, 37C60, 65K10.

1 Geometry and the selection theorem

Let $M \subset \mathbb{R}^d$ be a compact embedded C^5 manifold without boundary, and let $F \in C^5$ on an open neighborhood of M. Suppose, for every $p \in M$, that

$$\nabla F(p) = 0, \qquad \ker \nabla^2 F(p) = T_p M, \qquad \operatorname{spec}(H_p) \subset [\mu, L], \quad 0 < \mu \le L < 8, \tag{1}$$

where $H_p = \nabla^2 F(p)|_{N_p M}$. Denote the orthogonal tangent and normal projectors by P_p, N_p, and the nearest-point projection in a sufficiently small tube by π. Put

$$\begin{aligned} q(p) &= \tfrac{1}{2}|p|^2, \quad g(p) = \operatorname{grad}_M q(p) = P_p p, \\ \nu(p) &= N_p p, \quad \psi(p) = \tfrac{1}{2}\langle \nu(p), H_p^{-1}\nu(p)\rangle. \end{aligned} \tag{2}$$

Thus q is the primary potential and ψ is the secondary potential. The second fundamental form convention is $\mathrm{II}_p(a, b) = N_p D_a b$; it gives inward acceleration for geodesics on the unit sphere. Different connected components of M may have different values of F. Fix, on the original integer clock,

$$b = \tfrac{1}{4}, \qquad \gamma = \tfrac{3}{2}, \qquad h_k = k^{-1/3}, \qquad \varepsilon_k = h_k^2 = k^{-2/3}. \tag{3}$$

For $k \geq K$ the actual two-line recursion is

$$\begin{aligned} y_k &= x_k + (1 - \gamma h_k)(x_k - x_{k-1}) - b\big(\nabla F(x_k) - \nabla F(x_{k-1})\big), \\ x_{k+1} &= y_k - b\big(\nabla F(y_k) + h_k^2 y_k\big). \end{aligned} \tag{4}$$

The gradient-history coefficient in the first line is the same $b = 1/4$ as in the second line. This is the specialization $s = 1/4$, $\delta = 3$, $\beta = 1/2$ of the boxed IGATH-2 scheme in the final published version of [1, Section 4.2, p. 45]. That source supplies the algorithm; its convergence results are not used to prove the local nonconvex theorem below.

Use the actual current-base history variables

$$\begin{aligned} p_k &= \pi(x_k), & n_k &= x_k - p_k, \\ u_k &= N_{p_k}(x_k - x_{k-1}), & v_k &= h_k^{-1} P_{p_k}(x_k - x_{k-1}). \end{aligned} \tag{5}$$

In particular, $x_k - x_{k-1} = h_k v_k + u_k$ exactly. The normal history u_k is not $n_k - n_{k-1}$. A late proof suffix always keeps the coefficients indexed by k.

The question is whether this clock produces genuine secondary selection. Under the reciprocal schedule $h_k = k^{-1/2}$, the corresponding center increments $h_k^3 W$ are summable. Here $h_k^3 = 1/k$ is not summable, so finite center motion cannot be inferred from a size bound alone. Sections 2–7 establish the general mechanism, including the exact extra hypothesis needed for point convergence. Section 8 establishes a uniform circle basin and rate that do not follow from eventual general estimates. Section 9 compares the two schedules from the identical history and records the sharp upper normal endpoint.

The collision-safe metric below is the common normal preliminary also used in the reciprocal-clock companion [6]. The critical capture and all secondary-selection proofs are given here in full. Only the reciprocal comparison and the endpoint consequence use results from that companion.

Theorem 1 (Capture and secondary selection)**.** *Under* (1)*, there exist $r > 0$ and $\theta > 0$, depending only on the displayed geometry, with the following property. For every finite $B_0 \geq 0$ there is an original index $K_0(B_0)$ such that every $K \geq K_0(B_0)$ and independent history pair satisfying*

$$x_K \in \{x : \operatorname{dist}(x, M) < r\}, \qquad |n_K| \leq \theta h_K, \quad |u_K| \leq \theta h_K, \quad |v_K| \leq B_0 \tag{6}$$

satisfy the conclusions below. No earlier prefix is assumed.

A. New-clock capture. *The actual suffix is well defined; x_k, y_k remain in a controlled tube, and v_k is bounded. For some suitable fixed geometry-dependent $\rho \in (0,1)$ and a finite constant $C = C(B_0)$, uniform over all admitted histories and all original $K \geq K_0(B_0)$,*

$$|n_k| + |u_k| \leq C\rho^{k-K}(|n_K| + |u_K|) + Ch_k^2 \leq 2C\theta h_K \rho^{k-K} + Ch_k^2. \tag{7}$$

Moreover,

$$\sum_{k \geq K} h_k\big(|v_k|^2 + |g(p_k)|^2\big) < \infty, \qquad v_k \longrightarrow 0, \quad g(p_k) \longrightarrow 0. \tag{8}$$

The sequences x_k, y_k have the same nonempty compact connected cluster set $\Omega \subset \operatorname{Crit}_M q$.

B. Secondary reduction. *Suppose $q|_M$ is Morse–Bott: its critical set consists of compact critical submanifolds whose Hessian kernel is precisely their tangent space. Then one connected critical component S contains Ω, and $\operatorname{dist}_M(p_k, S) \to 0$. For all sufficiently late k, write*

$$p_k = \exp_{s_k}^M(\eta_k), \qquad \eta_k \in V_{s_k}, \qquad V_s = (T_sS)^\perp \cap T_sM,$$

where s_k is the intrinsic tubular projection onto S. Let w_k be parallel transport of v_k from p_k to s_k. Set

$$\begin{aligned} a(p) &= -H_p^{-1}\nu(p), \qquad Q_s = \operatorname{Hess}_M q(s), \\ \Xi(s) &= (Q_s|_{V_s})^{-1} P_{V_s} \operatorname{grad}_M \psi(s), \\ W(s) &= \tfrac{1}{6} \operatorname{grad}_S \psi(s). \end{aligned} \tag{9}$$

An inverse on the zero-dimensional bundle means $\Xi = 0$. On a late actual suffix,

$$\begin{aligned} \eta_k &= h_k^2 \Xi(s_k) + O(h_k^3), & w_k &= h_k^2 W(s_k) + O(h_k^3), \\ n_k &= -h_k^2 H_{p_k}^{-1} \nu(p_k) + O(h_k^4), & u_k &= O(h_k^4). \end{aligned} \tag{10}$$

In particular, as ambient vectors relative to the moving center,

$$x_k = s_k + h_k^2\big(a(s_k) + \Xi(s_k)\big) + O(h_k^3), \qquad y_k = s_k + h_k^2\big(a(s_k) + \Xi(s_k)\big) + O(h_k^3). \tag{11}$$

These conclusions precede any center convergence or finite-length assumption. The actual center and ambient history obey

$$\operatorname{Log}_{s_k}^S(s_{k+1}) = \frac{1}{6k} \operatorname{grad}_S \psi(s_k) + r_k, \qquad |r_k| \le Ck^{-4/3}, \tag{12}$$

$$x_k - x_{k-1} = h_k^3 W(s_k) + O(h_k^4). \tag{13}$$

Without any KL assumption, $\psi(s_k)$ converges, $\operatorname{grad}_S \psi(s_k) \to 0$, and $\Omega \subset \operatorname{Crit}_S \psi$. The same conclusions hold in the localized variant when a captured trajectory has a cluster point in a compact isolated Morse–Bott component S, without assumptions on other primary components.

C. Point selection and finite actual paths. *Assume in addition that $f = -\psi|_S$ has the local one-sided power KL property at every potential critical cluster point s_*. Specifically, there are a neighborhood U, constants $c, \delta > 0$, and $\vartheta_* \in [0, 1)$ such that*

$$|\operatorname{grad}_S f(s)| \ge c\big(f(s) - f(s_*)\big)^{\vartheta_*} \quad \text{if } s \in U, \quad 0 < f(s) - f(s_*) < \delta. \tag{14}$$

It suffices to assume this near the actual cluster set. The general concave KL hypothesis in Section 6.5 is an alternative sufficient assumption. Then there is one $p_* \in S$ with

$$\operatorname{grad}_S \psi(p_*) = 0, \qquad x_k \to p_*, \quad y_k \to p_*, \quad s_k \to p_*; \tag{15}$$

and both actual paths have finite length:

$$\sum_{k \ge K} |x_{k+1} - x_k| < \infty, \qquad \sum_{k \ge K} |y_{k+1} - y_k| < \infty. \tag{16}$$

All late-suffix O constants may depend on the admitted actual history and its original entry index. The parameter ρ is one suitable contraction rate, not an arbitrarily prescribed number in $(0,1)$. Constants are chosen on the fixed interval $[\mu, L] \subset (0,8)$; no uniformity is asserted as $\mu \downarrow 0$ or $L \uparrow 8$. The entry test puts x_{K-1} in the domain of F for sufficiently late K. The conclusion selects a critical point of ψ on the actual selected primary component; it does not identify a global maximum of ψ, a globally least-norm point of M, or a basin-independent limit.

The regularity assumptions make ψ at least C^3. They do not imply its analyticity, even if M is analytic. The secondary KL hypothesis is additional; it follows, for example, from real analyticity of the actual restricted f in an analytic structure compatible with S, by the local analytic gradient inequality [2, 3]. Analyticity of the manifold alone does not supply that property for F or H. No rate for the center is asserted beyond the explicit augmented-energy tail bound proved below.

2 Local geometry and sharp normal stability

Fix nested tubes whose closures lie in the domain of F. Compactness and C^5 regularity provide uniform Taylor constants for F, the nearest projection, projectors, short intrinsic geodesics, and tangent and normal transports. Tubes are shrunk before choosing the entry index. A bar means transport back to the current base. All transports follow unique short geodesics; no global frame is chosen.

For a normal vector ζ and nearby bases r, p, the basic estimates are

$$\nabla F(p+\zeta) = H_p\zeta + O(|\zeta|^2), \tag{17}$$

$$|P_p \nabla F(r+\zeta)| \le C\big(|r-p||\zeta| + |\zeta|^2\big), \qquad \zeta \in N_r M, \tag{18}$$

$$\exp_p(a) = p + a + \tfrac{1}{2}\,\mathrm{II}_p(a,a) + O(|a|^3), \qquad a \in T_pM. \tag{19}$$

The second estimate follows from $P_r H_r \zeta = 0$ and Lipschitz variation of P. If $a \in T_pM$, $c \in N_pM$, $\delta = \mathrm{Log}_p r$, and τ is tangent parallel transport from p to r, then

$$\tau^{-1} P_r(a+c) = a + O\big(|\delta||c| + |\delta|^2|a|\big). \tag{20}$$

For the tangent input, the ambient first derivative of parallel transport and of its projected comparison is the same $\mathrm{II}_p(\delta, a)$, so their difference is second order. The normal-input estimate follows from projector variation. Local bundle charts and a finite cover make the bounds uniform.

Since $\pi(p+\zeta) = p$ for small normal ζ, bounded derivatives give

$$\mathrm{dist}_M\big(p, \pi(p+a+\zeta)\big) \le C|a|, \tag{21}$$

$$\mathrm{Log}_p\, \pi(p+a+\zeta) = a + O\big(|\zeta||a| + |a|^2\big). \tag{22}$$

The normal distance of $p+a+\zeta$ from M is $O(|\zeta|+|a|^2)$. To check the sharper logarithmic formula, its left side vanishes at $a = 0$, its derivative in a there is $I + O(|\zeta|)$, and its second derivatives are bounded. No cancellation of a quadratic tangent term is required.

The exact clock gives

$$\frac{h_k}{h_{k+1}} = (1 + h_k^3)^{1/3} = 1 + \tfrac{1}{3}h_k^3 + O(h_k^6), \tag{23}$$

$$h_{k+1}^2 - h_k^2 = -\tfrac{2}{3}h_k^5 + O(h_k^8). \tag{24}$$

In particular, $\sum h_k^3$ diverges, whereas $\sum h_k^4$ and $\sum h_k^5$ converge. That distinction requires a fresh capture argument.

2.1 Input-only finite-history estimates

At one original index write $h = h_k$, $p = \pi(x)$, $n = x - p$, and $x - x_{\text{prev}} = hv + u$. Fix finite caps R, D and suppose only the input bounds

$$|v| \le R, \qquad |n| + |u| \le Dh. \tag{25}$$

For sufficiently small h, depending on these fixed caps and the geometry, x_{prev}, y, x_+ all lie in the larger tube. Thus their projections and transports are defined without assuming a next-state working cap.

Put $z = (n, u)$, $A_p = I - bH_p$, and

$$B_p = \begin{pmatrix} A_p & A_p^2 \\ A_p - I & A_p^2 \end{pmatrix}. \tag{26}$$

The literal recursion gives

$$\bar{z}_+ = B_p z + R_N, \qquad |R_N| \le C_{R,D}h^2, \tag{27}$$

$$\operatorname{Log}_p p_+ = hv + r_p, \qquad |r_p| \le C_{R,D}h^2, \tag{28}$$

$$\bar{v}_+ = v - h(\gamma v + bg(p)) + r_v, \qquad |r_v| \le C_{R,D}\big(h^2 + |z| + |z|^2/h\big). \tag{29}$$

Here is a direct verification. The predecessor is $p + n - u - hv$. First-order expansion yields

$$\nabla F(x) - \nabla F(x_{\text{prev}}) = H_p u + O(h^2).$$

Hence

$$y - p = hv + n + A_p u + O(h^2), \qquad x_+ - p = hv + A_p n + A_p^2 u + O(h^2).$$

The new base has normal displacement $O(h^2)$ from p, proving the first row of (27). The identity

$$u_+ = N_{p_+}\big[(p_+ - p) + (n_+ - n)\big]$$

with chord-normal component $O(h^2)$ proves its second row. Transport changes on these caps are $O(h^2)$.

For the tangent estimate retain the normal dependence. The predecessor and predictor have normal displacement $O(|z|+h^2)$ and base change $O(h)$, so (18) gives tangent gradient errors $O(h|z| + |z|^2 + h^3)$. Also $P_p y = g(p) + hv + O(h^2)$. Therefore

$$P_p(x_+ - x) = hv - \gamma h^2 v - bh^2 g(p) + O(h^3 + h|z| + |z|^2).$$

The normal increment is $O(|z|+h^2)$. Apply (20), divide by h_+, and use (23) to obtain (29). Projection gives (28). These computations use the unchanged gradient-history coefficient.

2.2 The collision-safe metric

Lemma 1 (Exact scalar Schur interval)**.** *For a real number a, both eigenvalues of*

$$B(a) = \begin{pmatrix} a & a^2 \\ a-1 & a^2 \end{pmatrix}$$

lie strictly inside the unit disk if and only if $-1 < a < 1$. With $a = 1 - H/4$, this is exactly $0 < H < 8$.

Proof. The characteristic polynomial is

$$r^2 - a(1+a)r + a^2.$$

For $-1 < a < 1$ and $a \neq 0$, its discriminant is $a^2(a-1)(a+3) < 0$. Its conjugate roots therefore have modulus $|a| < 1$. At $a = 0$,

$$B(0) = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \qquad B(0)^2 = 0.$$

This nonzero nilpotent matrix is defective but stable. If $|a| \geq 1$, the product of the eigenvalues is $a^2 \geq 1$, so both cannot have modulus below one. At $a = -1$ the eigenvalues are $\pm i$; at $a = 1$ both are 1. ∎

Lemma 2 (Collision-safe metric)**.** *On $-1 < a < 1$, set*

$$R(a) = 2a^2 + a + 1, \qquad D(a) = (1-a^2)R(a)$$

and define a symmetric matrix $Q(a)$ by

$$\begin{aligned} Q_{11}(a) &= \frac{2 + 3a^2 - a^3 - a^4 - a^5}{D(a)}, \\ Q_{12}(a) &= \frac{a^2(a^3 + a^2 + 2a - 1)}{D(a)}, \\ Q_{22}(a) &= \frac{a^6 + 2a^4 - 2a^3 + 1}{(1-a)D(a)}. \end{aligned} \tag{30}$$

Then $Q(a) > 0$, $Q(a) \geq I$, and

$$Q(a) - B(a)^* Q(a) B(a) = I, \qquad Q(0) = \mathrm{diag}(2, 1). \tag{31}$$

Rational functional calculus defines a positive metric $\mathcal{Q}_p$ on $N_pM \oplus N_pM$ with

$$I \leq \mathcal{Q}_p \leq M_Q I, \qquad B_p^* \mathcal{Q}_p B_p = \mathcal{Q}_p - I \tag{32}$$

for one finite $M_Q > 1$. The matrix function Q is analytic on its spectral domain; the bundle metric $\mathcal{Q}_p$ inherits the finite regularity of H_p, at least C^3 under the stated C^5 data. In particular all derivatives used below are uniformly bounded.

Proof. Every denominator is nonzero because

$$R(a) = 2(a + 1/4)^2 + 7/8 > 0.$$

Direct polynomial multiplication verifies (31). Positivity can be checked without presupposing stability:

$$Q_{22}(a) = \frac{(a^3 - 1)^2 + 2a^4}{(1 - a)D(a)} > 0,$$

$$\det Q(a) = \frac{a^4 + (a^2 - a)^2 + 2(1 - a)}{(1 - a)^3 (1 + a)^2 R(a)} > 0.$$

The two conditions imply positive definiteness. The Lyapunov identity then implies $Q(a) \geq I$.

In (30), replace a by A_p and scalar constants by multiples of the identity. The denominator operators $I - A_p$, $I - A_p^2$, and $I + A_p + 2A_p^2$ are uniformly invertible on the fixed compact spectral interval. Orthogonal diagonalization at a single p proves positivity and (32), because all blocks commute. No eigenbasis is required to vary with p. In local orthonormal bundle frames, matrix inversion and multiplication preserve the available finite differentiability. Compactness gives the uniform derivative bounds. Thus eigenvalue crossings, repeated eigenvalues, and the defective $A_p = 0$ block require no additional hypothesis. ∎

The identity gives the uniform frozen-power bound

$$\|B_p^j\| \leq \sqrt{M_Q}(1 - 1/M_Q)^{j/2}, \qquad j \geq 0.$$

A common Euclidean one-step contraction and stability under arbitrary switching are not asserted. No inverse of A_p or B_p is used. On the input caps the base moves by $O(h)$, so the transported metric varies by $O(h)$.

Choose one $\rho \in (0, 1)$ above the frozen metric contraction factor. For sufficiently small h,

$$|z_+|_{\mathcal{Q}_{p_+}} \leq \rho |z|_{\mathcal{Q}_p} + C_{R,D} h^2. \tag{33}$$

On every interval where the input caps hold, iteration gives

$$|z_k| \le C_0\theta h_K\rho^{k-K} + C_1h_k^2. \tag{34}$$

The homogeneous coefficient depends only on metric equivalence and the two entry bounds; C_1 can depend on R, D. For $i \le k-1$ and $\ell = k-1-i$,

$$\frac{h_i^2}{h_k^2} = (k/i)^{2/3} \le (\ell+2)^{2/3}, \qquad \sup_{j\ge0}\frac{h_K}{h_{K+j}}\rho^j \le \sup_{j\ge0}(1+j)^{1/3}\rho^j < \infty.$$

These bounds prove the convolution estimate on the original clock. Before inserting the entry bound, the same iteration retains the sharper homogeneous term $C\rho^{k-K}|z_K|$, which gives the first inequality in (7). Choose D large enough that its quarter-cap dominates the homogeneous normal-cap bound and the entry. After R, D are fixed, increasing K makes $C_1h_k^2$ improve the total cap to $Dh_k/2$.

Finally, substitute (34) and $|z|^2/h \le D|z|$ into (29). With fixed constants,

$$|r_{v,k}| \le C_2h_k^2 + t_k, \qquad t_k = C_3\theta h_K\rho^{k-K}, \qquad \sum_{k\ge K} t_k \le C_4\theta h_K. \tag{35}$$

The actual entry transient is retained; it need not be $O(h_K^2)$.

3 Critical-clock capture and primary dissipation

Let $G = \max_M |g|$ and define

$$B_* = \max\{B_0, bG/\gamma + 1\}, \qquad R = B_* + 2.$$

Choose D as in Section 2.2, after fixing this R. All coarse Taylor and contraction constants are now fixed. Choose the original K_0 sufficiently large for the geometry, normal-cap improvement, $\gamma h < 1$, and

$$C_2h \le \gamma, \qquad \sum_{k\ge K} t_k < \tfrac{1}{2}, \tag{36}$$

together with the auxiliary-cap requirements below. These are finitely many smallness conditions on h_K, improving for every $K \ge K_0$.

While the input caps hold, (29) and (35) imply

$$\begin{aligned} |v_{k+1}| &\le (1-\gamma h_k)|v_k| + bGh_k + C_2h_k^2 + t_k \\ &\le (1-\gamma h_k)|v_k| + \gamma h_kB_* + t_k. \end{aligned}$$

Induction gives the cumulative-transient barrier

$$|v_{k+1}| \le B_* + \sum_{j=K}^{k} t_j < B_* + \tfrac{1}{2} < R - 1. \tag{37}$$

No summation of h^2 or h^3 is used.

The argument applies at a prospective first output violation. The raw input-only normal estimate first puts the next scaled state into a fixed larger auxiliary normal cap; the raw velocity estimate gives $|v_+| < R + 1$ for late K. Equation (33) applies to that constructed step because its metric estimate uses the input-derived $O(h)$ base change, not a next working cap. Iteration through that step improves the next normal cap to $D/2$, and (37) improves the next velocity cap to $R - 1$. The new iterate and predictor normal distances are then strictly inside the chosen tube. Thus no first violation occurs.

At entry, the predecessor lies within $(2\theta + B_0)h_K$ of p_K, so it is in the domain of F. All gradient evaluations and projections have therefore been justified in their order of construction. This proves capture and (7), with a constant $C = C(B_0)$ once the geometry is fixed. The choice of $\theta > 0$ can be fixed from the geometry before B_0: its size is absorbed by the subsequent fixed D and late $K_0(B_0)$, with no circular choice of a forcing constant.

For each fixed admitted history, the geometric transient is eventually smaller than every power of k. Hence at a history-dependent late index J,

$$|n_k| + |u_k| \le Ch_k^2 \quad (k \ge J). \tag{38}$$

This later suffix suppresses the transient without restarting the integer clock.

3.1 Primary dissipation with a summable fifth-order error

Work on the bounded velocity cap and the regime (38). The predecessor has current-base tangent coordinate $-hv$ and normal coordinate $n - u$. Equations (18) and (21) yield

$$|P_p \nabla F(x)| \le Ch^4, \qquad |P_p \nabla F(x_{\mathrm{prev}})| \le C(h^3|v| + h^4).$$

Thus the literal predictor satisfies

$$P_p(y - p) = hv - \gamma h^2 v + O(h^3|v| + h^4), \qquad N_p(y - p) = O(h^2).$$

Its projected base is $O(h|v| + h^4)$ from p, and its normal displacement is $O(h^2)$. A further use of (18) gives $|P_p \nabla F(y)| \le C(h^3|v| + h^4)$. With $\alpha = P_p(x_+ - x)$ and $c = N_p(x_+ - x)$,

$$\alpha = hv - \gamma h^2 v - bh^2 g(p) + O(h^3|v| + h^4), \qquad |c| \le Ch^2. \tag{39}$$

Apply (22) to $x_+ = p + \alpha + (n + c)$, and (20) to the actual new history. Then

$$\mathrm{Log}_p p_+ = hv + r_p, \qquad |r_p| \le C\big[h^2(|v| + |g|) + h^4\big], \tag{40}$$

$$\bar{v}_+ = v - h(\gamma v + bg) + r_v, \qquad |r_v| \le C\big[h^2(|v| + |g|) + h^3\big]. \tag{41}$$

The normalization uses exactly (23); its error $O(h^3|v|)$ is included in (41). We also retain

$$\mathrm{dist}_M(p_+, p) \le C\big[h|v| + h^2|g| + h^4\big]. \tag{42}$$

Let $D_q = \max_M |\operatorname{Hess}_M q|$. Choose $0 < \epsilon \le 1$ with $\epsilon(D_q + 9/2) \le 3/4$, and define on the capped tangent bundle

$$\mathcal{V}(p, v) = \tfrac{1}{2}|v|^2 + bq(p) + \epsilon\langle g(p), v\rangle. \tag{43}$$

For the intrinsic vector $(v, -\gamma v - bg)$, direct differentiation gives

$$\begin{aligned} D\mathcal{V} &= -\gamma|v|^2 + \epsilon \operatorname{Hess}_M q[v, v] - \epsilon\gamma\langle g, v\rangle - \epsilon b|g|^2 \\ &\le -\tfrac{3}{4}|v|^2 - \tfrac{\epsilon}{8}|g|^2. \end{aligned}$$

Here $\gamma|g||v| \le (b/2)|g|^2 + (\gamma^2/(2b))|v|^2$ and $\gamma^2/(2b) = 9/2$. This is a first-derivative identity, not replacement of the discrete recursion by an ODE.

Write $A_0 = |v| + |g|$. The horizontal and vertical first derivatives of $\mathcal{V}$ are bounded by CA_0, and its second derivative is bounded on the cap. Taylor expansion at the actual increments (40)–(41) gives

$$\Delta\mathcal{V} \le -chA_0^2 + Ch^2A_0^2 + Ch^3A_0 + Ch^6.$$

Indeed the leading phase increment is $O(hA_0)$, the defect is $O(h^2A_0 + h^3)$, and the first derivatives vanish linearly with A_0. Absorb $Ch^2A_0^2$ for late h and use Young's inequality on Ch^3A_0. The resulting actual energy estimate is

$$\mathcal{V}_{k+1} - \mathcal{V}_k \le -c_1h_k\big(|v_k|^2 + |g(p_k)|^2\big) + C_5h_k^5. \tag{44}$$

Its error is summable on this clock. Since $\mathcal{V} \ge |v|^2/4 - \epsilon^2G^2$, summation proves the series in (8), first from J and then from K by adding the finite prefix.

Put $a_k = |v_k|^2 + |g(p_k)|^2$. The raw phase bounds give $|a_{k+1} - a_k| \le Ch_k$. If $a_k \ge \delta > 0$ infinitely often, each such excursion maintains $a_j \ge \delta/2$ over a fixed positive length of the divergent clock $\sum h_k$. Disjoint excursions would each contribute a fixed positive amount to $\sum h_ka_k$, a contradiction. Individual clock steps tend to zero, so endpoint errors do not change the argument. Thus $v_k, g(p_k) \to 0$.

Compactness gives $\operatorname{dist}(p_k, \operatorname{Crit}_M q) \to 0$. The same holds for x_k because $n_k \to 0$. The actual history $d_k = h_kv_k + u_k$ tends to zero. Bounded Hessian on the controlled compact neighborhood gives $|y_k - x_k| \le C|d_k|$, so the two sequences have the same nonempty compact cluster set $\Omega \subset \operatorname{Crit}_M q$. It is connected: if it split into two disjoint compact pieces, all late points would lie in their separated neighborhoods, while vanishing steps would prevent passage between those neighborhoods. Both pieces could not then contain cluster points. This proves part A without Morse–Bott or KL assumptions.

4 Localization and the full moving phase bundle

Suppose q is Morse–Bott. Each compact connected critical component S is locally isolated within $\operatorname{Crit}_M q$. In coordinates along and transverse to S, the transverse derivative

of g is the invertible transverse Hessian. The implicit function theorem identifies its local zero set with S; every full zero of g lies there. A finite cover makes the isolation uniform. This description also gives S the needed C^4 regularity, because g is C^4 in C^5 coordinates. Compactness of $\mathrm{Crit}_M q$ and its locally isolating component neighborhoods imply that only finitely many components occur. Connectedness of Ω therefore places it in one S.

In the localized variant, a compact isolated component is relatively open and closed in $\mathrm{Crit}_M q$, so a connected Ω meeting it lies in it. In either case,

$$\mathrm{dist}_M(p_k, S) \to 0, \qquad v_k \to 0. \tag{45}$$

No point convergence or finite motion inside S has been shown.

At late indices use the tubular variables of Theorem 1 and set

$$Z_k = (\eta_k, w_k) \in E_{s_k}, \qquad E_s = V_s \oplus T_s M. \tag{46}$$

Then $Z_k \to 0$ and $|Z_k|$ is comparable to $\mathrm{dist}_M(p_k, S) + |v_k|$. The Hessian Q_s kills $T_s S$ and is invertible on V_s. Define

$$L_s(\eta, w) = \big(P_{V_s} w, -bQ_s\eta - \gamma w\big). \tag{47}$$

All velocity modes are retained. Each nonzero transverse eigenvalue λ of Q_s gives roots of $r^2 + \gamma r + b\lambda = 0$. Positive λ gives stable roots; negative λ gives one stable and one unstable root. Every velocity direction tangent to S gives the stable root $-\gamma$. Thus L_s is uniformly hyperbolic on compact S; only equilibrium-position centers have been removed.

Let $T_k : E_{s_k} \to E_{s_{k+1}}$ be connection transport, using the normal connection of S in M on V and the Levi-Civita connection of M along the short geodesic of S on TM. The finite Taylor expansion of the actual map (40)–(41) yields

$$T_k^{-1} Z_{k+1} = (I + h_k L_{s_k}) Z_k + R_k, \qquad |R_k| \le C\big[h_k |Z_k|^2 + h_k^2 |Z_k| + h_k^3\big]. \tag{48}$$

Its linear part follows from the already derived first-order phase increment $h(v, -\gamma v - bg)$. At $\eta = w = 0$, the base derivative is $P_{T_s S} w$ and the fiber derivative is exactly (47). Coordinate and connection variations multiply at least two phase variables, contributing $O(h|Z|^2)$; raw defects and finite-step Taylor remainders supply the other terms.

Independently, projection of (42) gives

$$d_k^S := \mathrm{dist}_S(s_{k+1}, s_k) \le C\big[h_k |Z_k| + h_k^4\big] = o(h_k). \tag{49}$$

This estimate is established before constructing the dichotomy. Its sum need not be finite.

4.1 Spectral subbundles and splitting-preserving transport

Let $P_s^{\mathrm{st}}, P_s^{\mathrm{un}}$ be the stable and unstable spectral projectors of L_s. Separation of the spectral regions gives projectors with the needed bounded derivatives, using local resolvent

contours agreeing on overlaps. No choice of individual eigenvectors is needed. Repeated or defective eigenvalues within a block cause no difficulty.

On the stable subbundle use the Lyapunov form

$$\mathcal{Q}_s^{\mathrm{st}}(z,z) = \int_0^\infty |e^{tL_s} z|^2 \, dt,$$

and on the unstable subbundle the analogous form with e^{-tL_s}. Compactness and finite-dimensional resolvent contours give uniform exponential bounds; polynomial Jordan factors are absorbed by a smaller exponential rate. The forms are uniformly equivalent to the original norms and have bounded first derivatives. Their Lyapunov identities give, for small h and fixed $a_0 > 0$,

$$\begin{aligned} |(I + hL_s) z_{\mathrm{st}}|_{\mathcal{Q}_s^{\mathrm{st}}} &\le (1 - 2a_0 h) |z_{\mathrm{st}}|_{\mathcal{Q}_s^{\mathrm{st}}}, \\ |(I + hL_s)^{-1} z_{\mathrm{un}}|_{\mathcal{Q}_s^{\mathrm{un}}} &\le (1 - 2a_0 h) |z_{\mathrm{un}}|_{\mathcal{Q}_s^{\mathrm{un}}}. \end{aligned}$$

Correct connection transport to preserve the splitting:

$$D_k = P^{\mathrm{st}}_{s_{k+1}} T_k P^{\mathrm{st}}_{s_k} + P^{\mathrm{un}}_{s_{k+1}} T_k P^{\mathrm{un}}_{s_k}. \tag{50}$$

Then $|D_k - T_k| \le C d_k^S$. For nearby bases D_k is invertible and block preserving; it and its inverse change the corresponding varying Lyapunov norms by factors at most $1 + C d_k^S$. Put

$$A_k = D_k (I + h_k L_{s_k}), \qquad t_k = \sum_{m=J}^{k-1} h_m.$$

Since $d_k^S = o(h_k)$, a late suffix has block contraction and inverse-block contraction by $1 - a_0 h_k$. With the associated evolution products $\Phi(k,l)$,

$$\begin{aligned} |\Phi(k,l) P^{\mathrm{st}}_{s_l}| &\le C e^{-a(t_k - t_l)}, \quad k \ge l, \\ |\Phi(k,l) P^{\mathrm{un}}_{s_l}| &\le C e^{-a(t_l - t_k)}, \quad k \le l. \end{aligned} \tag{51}$$

This proves a dichotomy along the actual moving base. Uniform hyperbolicity under arbitrary switching is not assumed.

4.2 A two-space Green argument before center convergence

Using (49), rewrite the exact recurrence along the actual trajectory as

$$Z_{k+1} = A_k Z_k + h_k E_k Z_k + r_k, \qquad |E_k| \le C(|Z_k| + h_k) \to 0, \qquad |r_k| \le C h_k^3. \tag{52}$$

For precision, split a vector bounded by $C[h|Z|^2 + h^2|Z| + h^3]$ into one bounded by $Ch(|Z| + h)|Z|$ and one bounded by Ch^3. When $Z \neq 0$, represent the first as hEZ by a rank-one operator; set $E = 0$ at $Z = 0$. The correction $(T - D)(I + hL)Z$ has the same

bound by (49). These coefficient sequences are now frozen along the existing trajectory; no nonlinear uniqueness is presumed.

For forcing f_j in the next fiber, define

$$\begin{aligned}(\mathcal{G}f)_k = &\sum_{j=J}^{k-1} \Phi(k, j+1) P^{\mathrm{st}}_{s_{j+1}} f_j \\ &- \sum_{j=k}^{\infty} \Phi(k, j+1) P^{\mathrm{un}}_{s_{j+1}} f_j.\end{aligned} \tag{53}$$

Every bounded solution of $U_{k+1} = A_k U_k + f_k$ has

$$U_k = \Phi(k, J) P^{\mathrm{st}}_{s_J} U_J + (\mathcal{G}f)_k. \tag{54}$$

The stable formula is forward variation of constants. For the unstable part, backward iteration has a terminal term tending to zero because U is bounded and $\sum h_k = \infty$. It is not an extra condition on the history.

Let X_∞ be the bounded sections with supremum norm and X_m the sections with norm $\sup_k |U_k|/h_k^m$. The original-clock weight comparison for $j < k$ is

$$\log \frac{h_j^m}{h_k^m} = \frac{m}{3} \log \frac{k}{j} \le \frac{m}{3} \sum_{\ell=j}^{k-1} \frac{1}{\ell} \le \frac{m}{3} h_J^2 (t_k - t_j). \tag{55}$$

For $j \ge k$, the weight ratio is at most one. Choose J late enough that $(m/3)h_J^2$ consumes at most half the kernel exponent. Comparing with integrals over consecutive slow-clock intervals shows that $\sum_j h_j e^{-\beta|t_k - t_j|}$ is uniformly bounded. Consequently,

$$\begin{aligned}\|\mathcal{G}(hEU)\|_m &\le C_G \sup_j |E_j| \, \|U\|_m, \\ \|\mathcal{G}r\|_m &\le C_G C_r \quad \text{if } |r_k| \le C_r h_k^{m+1}.\end{aligned} \tag{56}$$

These bounds hold on X_∞ as well, taking $m = 0$. The stable homogeneous term belongs to every fixed X_m by the same comparison.

Fix kernel constants on a preliminary late suffix, then increase J until $C_G \sup |E| < 1/2$. The affine fixed-point equation for (52) is a contraction on both X_∞ and X_2. Its X_2 fixed point is also in X_∞; uniqueness in X_∞ identifies it with the actual bounded solution. Therefore

$$|Z_k| = O(h_k^2), \qquad \mathrm{dist}_M(p_k, S) + |v_k| + |g(p_k)| = O(h_k^2). \tag{57}$$

This retains every velocity mode and proves the rates before any center point limit or finite center path. No possibly infinite weighted norm of the actual sequence has been absorbed.

5 Normal filtering and the secondary force

Equations (38) and (57) give

$$n, u = O(h^2), \qquad v, g = O(h^2), \qquad \operatorname{Log}_p p_+ = O(h^3).$$

Expand the literal normal equations at p. The gradient-history difference is $H_p u + O(h^4)$: the actual increment has size $O(h^2)$ and its nonlinear gradient difference is $O((|n|+|d|)|d|)$. The predictor normal coordinate is $n + (A_p - \gamma h)u + O(h^4)$. Thus

$$\bar{z}_+ = B_p z - bh^2\big(\nu(p), \nu(p)\big) + O(h^3). \tag{58}$$

The largest omitted term is the damping contribution $-\gamma h A_p u = O(h^3)$. Quadratic normal-gradient and regularization-offset terms are $O(h^4)$; projection and normal-transport changes are smaller because the base step is $O(h^3)$. In the current-base history row, $\bar{u}_+ = \bar{n}_+ - n$ up to chord-normal and transport errors within $O(h^3)$.

Set $a(p) = -H_p^{-1}\nu(p)$. The exact algebraic identity is

$$(a, 0) = B_p(a, 0) - b(\nu, \nu). \tag{59}$$

This target is not asserted to be invariant. Its actual variation obeys

$$\overline{h_+^2(a(p_+), 0)} - h^2(a(p), 0) = O(h^5), \tag{60}$$

by (24), bounded derivatives, and the $O(h^3)$ base step. Subtract it from (58) and apply the fast metric contraction. The forcing is $O(h^3)$; geometric convolution with the original polynomial weights gives

$$n = h^2 a(p) + O(h^3), \qquad u = O(h^3). \tag{61}$$

In particular $d = hv + u = O(h^3)$. This is filtering of the actual history, not a prepared regularized branch.

5.1 The actual secondary force

Define the tangent vector

$$\mathcal{T}(p) = P_p D^3 F(p)[a(p), a(p)].$$

Under (57) and (61), Taylor expansion of the gradient difference itself gives

$$\nabla F(x) - \nabla F(x - d) = H_p u + O\big((|n| + |d|)|d|\big) = H_p u + O(h^5).$$

The literal predictor therefore satisfies

$$\begin{aligned} P_p(y - p) &= hv - \gamma h^2 v + O(h^5), \\ N_p(y - p) &= n + (A_p - \gamma h)u + O(h^5) = h^2 a(p) + O(h^3). \end{aligned} \tag{62}$$

Its tangent displacement is $O(h^3)$. The quadratic normal term is the only order-h^4 contribution to its tangent gradient:

$$P_p \nabla F(y) = \tfrac{1}{2} h^4 \mathcal{T}(p) + O(h^5). \tag{63}$$

Indeed tangent-normal cross products are $O(h^5)$, the square of the tangent displacement is $O(h^6)$, and the third-order gradient remainder is $O(h^6)$. The literal regularizer is evaluated at y, so

$$h^2 P_p y = h^2 g(p) + O(h^5). \tag{64}$$

Consequently,

$$\alpha = P_p(x_+ - x) = hv - \gamma h^2 v - b h^2 g(p) - \tfrac{b}{2} h^4 \mathcal{T}(p) + O(h^5). \tag{65}$$

The normal increment c is $O(h^3)$: the apparent order-h^2 force $-bH_p n - bh^2 \nu(p)$ is $O(h^3)$, as are the history terms. Since $n + c = O(h^2)$ and $\alpha = O(h^3)$, (22) gives $\operatorname{Log}_p p_+ = \alpha + O(h^5)$. The transport error in (20) is $O(h^6)$. Dividing by the exact h_+ gives

$$\bar{v}_+ = v - h(\gamma v + b g(p)) - \tfrac{b}{2} h^3 \mathcal{T}(p) + O(h^4), \tag{66}$$

$$\operatorname{Log}_p p_+ = hv + O(h^4). \tag{67}$$

The clock normalization correction is $O(h^3 v) = O(h^5)$; it is not another order-h^3 forcing.

At every primary critical point $s \in S$,

$$\operatorname{grad}_M \psi(s) = -\tfrac{1}{2} \mathcal{T}(s). \tag{68}$$

To prove this, differentiate $p = g(p) + \nu(p)$ along $t \in T_p M$ and take the ambient normal component. The derivative of the identity map is tangent, so

$$\nabla^\perp_t \nu = -\operatorname{II}_p(t, g(p)).$$

At s, this derivative vanishes for every $t \in T_s M$, including directions transverse to S within M. Use normal vector fields parallel for the normal connection at s. Their ambient derivatives are tangent, and $\nabla^2 F(s)$ kills tangent vectors; hence

$$\langle a, (\nabla^\perp_t H) a \rangle = D^3 F(s)[t, a, a].$$

Differentiating $\psi = \langle \nu, H^{-1} \nu \rangle / 2$ now yields

$$d\psi_s[t] = -\tfrac{1}{2} D^3 F(s)[t, a, a],$$

which is (68). This identifies the normal-stiffness force; it cannot be read from q alone, which is constant on S.

5.2 Refined moving coefficients without a center limit

Transform (66)–(67) into the tubular phase coordinates of Section 4. Since $|Z| = O(h^2)$, transport of g to s gives $Q_s\eta + O(|\eta|^2)$ and transport of $\mathcal{T}(p)$ gives $\mathcal{T}(s) + O(|\eta|)$. By (68), the actual recurrence is

$$T_k^{-1} Z_{k+1} = (I + h_k L_{s_k}) Z_k + h_k^3 C(s_k) + O(h_k^4), \qquad C(s) = (0, b \operatorname{grad}_M \psi(s)). \tag{69}$$

There is no phase-independent order-h^3 forcing in the position equation. Coordinate corrections beyond the linear part are $O(h|Z|^2) = O(h^5)$; moving-frame corrections multiply a base step $O(h^3)$ and a fiber $O(h^2)$, so are also $O(h^5)$. All bounds are uniform on compact S without a limiting center.

The unique algebraic section solving $L_s\zeta + C(s) = 0$ is

$$\zeta(s) = (\Xi(s), W(s)), \qquad \Xi = (Q_s|_{V_s})^{-1} P_{V_s} \operatorname{grad}_M \psi, \quad W = \frac{b}{\gamma} \operatorname{grad}_S \psi = \frac{1}{6} \operatorname{grad}_S \psi. \tag{70}$$

The first row requires $P_V W = 0$, the transverse second row is $Q_s \Xi = P_V \operatorname{grad}_M \psi$, and the center-velocity second row is $\gamma W = b P_{T_s S} \operatorname{grad}_M \psi$. This algebra alone does not yet identify an actual solution.

Define $e_k = Z_k - h_k^2 \zeta(s_k)$. By (49) and (57), $d_k^S = O(h_k^3)$. Bounded derivatives and (24) imply

$$T_k^{-1} \left[h_{k+1}^2 \zeta(s_{k+1}) \right] - h_k^2 \zeta(s_k) = O(h_k^5). \tag{71}$$

Subtracting this target from (69) gives forcing $O(h^4)$. Replacing T_k by the splitting-preserving D_k costs $O(d_k^S |e_k|) = O(h_k^5)$, because $e_k = O(h_k^2)$ is already known. Thus the actual bounded section satisfies

$$e_{k+1} = A_k e_k + f_k, \qquad |f_k| \le C h_k^4. \tag{72}$$

Apply (54) and (55) with $m = 3$. The backward unstable terminal term vanishes by boundedness, the Green forcing is $O(h_k^3)$, and the stable homogeneous term is $O(h_k^3)$. Therefore

$$\eta_k = h_k^2 \Xi(s_k) + O(h_k^3), \qquad w_k = h_k^2 W(s_k) + O(h_k^3). \tag{73}$$

All full velocity directions, stable and unstable transverse phase modes, and moving-base corrections have been included before assuming convergence inside S.

5.3 Second normal filtering and the harmonic center equation

Repeat the literal normal calculation after (61). Now $n = h^2 a + O(h^3)$, $u = O(h^3)$, $d = O(h^3)$, and the base step is $O(h^3)$. The nonlinear gradient-history difference is $O(h^5)$, the quadratic normal-gradient term is $O(h^4)$, and the damping history term is $O(h^4)$. Hence

$$\bar{z}_+ = B_p z - b h^2 (\nu(p), \nu(p)) + O(h^4). \tag{74}$$

The exact new-base normal history differs from $\bar{n}_+ - n$ by a chord-normal term $O(h^6)$ and transport changes $O(h^5)$, so the second row has the same forcing. Subtract the target in (59), whose stepwise variation remains $O(h^5)$. Fixed fast contraction with $O(h^4)$ forcing gives

$$n_k = h_k^2 a(p_k) + O(h_k^4), \qquad u_k = O(h_k^4). \tag{75}$$

This estimate is essential for an actual finite-path conclusion. A normal-history remainder only of order h_k^3 would not be summable on this clock.

The derivative of the intrinsic center projection at $\eta = 0$ is P_{T_sS}. Equations (67), (73), and $|\eta| = O(h^2)$ give

$$\begin{aligned} \mathrm{Log}^S_{s_k}(s_{k+1}) &= h_k P_{T_{s_k}S} w_k + O(h_k^4) \\ &= h_k^3 W(s_k) + O(h_k^4) = \frac{1}{6k} \operatorname{grad}_S \psi(s_k) + r_k, \qquad |r_k| \le C k^{-4/3}. \end{aligned} \tag{76}$$

Replacing the coordinate derivative at η by its value at zero costs $O(|\eta|)$ times the actual phase step $O(h^3)$, hence $O(h^5)$. The coefficient error in (73), multiplied by h, is $O(h^4)$; finite-step coordinate terms are smaller. No summation of h^3 has been used.

Finally, the exact history decomposition and (75) give

$$x_k - x_{k-1} = h_k^3 W(s_k) + O(h_k^4) \tag{77}$$

as an ambient-vector equality. Transport from s_k to p_k changes a vector of size h_k^2 by $O(|\eta_k| h_k^2) = O(h_k^4)$; multiplication by h_k makes its contribution $O(h_k^5)$. This verifies the ambient interpretation. Finally $p_k = s_k + \eta_k + O(|\eta_k|^2)$, $a(p_k) = a(s_k) + O(|\eta_k|)$, and the normal estimate give the first equality of (11). Bounded Hessian in the literal predictor yields $|y_k - x_k| \le C|x_k - x_{k-1}| = O(h_k^3)$ and hence its second equality. These are moving-center expansions, not fixed-limit second-order coefficients.

6 A deterministic KL finite-length lemma

The next lemma is stated at the finite regularity actually needed above. A compact C^4 manifold with a C^3 Riemannian metric has the short logarithms, positive uniform injectivity radius, metric-compatible transports, and bounded second-order geodesic geometry used in the proof. Only two derivatives of the function are required. All notation in this section is local to the lemma.

Lemma 3 (Finite length under secondary KL)**.** *Let S be a compact C^4 Riemannian manifold without boundary, with a C^3 metric, and let $f \in C^2(S)$. Put $h_k = k^{-1/3}$, $b_k = h_k^3 = k^{-1}$, and $a_k = b_k/6$. Suppose that, for all sufficiently large k,*

$$u_k := \mathrm{Log}_{s_k}(s_{k+1}) = -a_k \operatorname{grad} f(s_k) + r_k, \qquad \|r_k\| \le R h_k^4. \tag{K1}$$

The logarithm is the short one, so $d_S(s_k, s_{k+1}) = \|u_k\|$. Equivalently one may assume the corresponding exponential recurrence: bounded gradient and positive injectivity radius make its steps short for all sufficiently large k.

Assume the following local one-sided power KL property at every critical point p that can be a cluster point of this sequence. There exist a neighborhood U_p, constants $\eta_p, \mu_p > 0$, and $\theta_p \in [0, 1)$ such that

$$\|\operatorname{grad} f(s)\| \geq \mu_p (f(s) - f(p))^{\theta_p} \quad \text{if } s \in U_p, \quad 0 < f(s) - f(p) < \eta_p. \tag{K2}$$

It is sufficient to assume this at every critical point of f. Then $f(s_k) \to \ell$, $\|\operatorname{grad} f(s_k)\| \to 0$,

$$\sum_k d_S(s_k, s_{k+1}) < \infty, \qquad s_k \longrightarrow s_\infty \in \operatorname{Crit} f, \qquad f(s_\infty) = \ell.$$

The value convergence and criticality hold even without KL. In the power case the explicit tail estimate (K17) *holds. The general concave KL hypothesis in Section 6.5 can replace* (K2)*, with tail estimate* (K21)*. In both cases the proof also gives $\sum_k h_k^3 \|\operatorname{grad} f(s_k)\| < \infty$.*

Proof. The finitely many indices before (K1) holds do not affect the conclusions. We work in one connected component of S.

6.1 Error absorption and a decreasing augmented energy

Write $g_k = \|\operatorname{grad} f(s_k)\|$. Let L bound the operator norm of the Riemannian Hessian of f on S. The geodesic Taylor inequality gives

$$f(s_{k+1}) \leq f(s_k) + \langle \operatorname{grad} f(s_k), u_k \rangle + \frac{L}{2}\|u_k\|^2.$$

Young's inequality and (K1) imply

$$\begin{aligned} \langle \operatorname{grad} f(s_k), r_k \rangle &\leq \frac{a_k}{4} g_k^2 + \frac{\|r_k\|^2}{a_k}, \\ \frac{L}{2}\|u_k\|^2 &\leq L a_k^2 g_k^2 + L\|r_k\|^2. \end{aligned}$$

Choose the initial index large enough that $La_k \leq 1/4$. Then

$$\begin{aligned} f(s_{k+1}) &\leq f(s_k) - \frac{a_k}{2} g_k^2 + \frac{5}{4a_k}\|r_k\|^2 \\ &\leq f(s_k) - \frac{1}{12} h_k^3 g_k^2 + 8R^2 h_k^5. \end{aligned} \tag{K3}$$

The numerical constant 8 merely rounds up $15/2$. In particular, the error cost is $O(h_k^5)$, not $O(h_k^4)$; it comes from $\|r_k\|^2/a_k$.

For every $k \geq 1$,

$$
\begin{aligned}
h_k^2 - h_{k+1}^2 &= \frac{2}{3}\int_k^{k+1} t^{-5/3}\,dt \geq \beta h_k^5, \qquad \beta := \frac{2}{3}\,2^{-5/3},\\
\gamma_0 h_k^4 \leq h_k - h_{k+1} &= \frac{1}{3}\int_k^{k+1} t^{-4/3}\,dt \leq \frac{1}{3}h_k^4, \qquad \gamma_0 := \frac{1}{3}\,2^{-4/3}.
\end{aligned}
\tag{K4}
$$

Fix, for example,

$$
A = \frac{8R^2+1}{\beta} > 0, \qquad E_k = f(s_k) + Ah_k^2.
$$

Equations (K3)–(K4) yield the decisive bound

$$
E_k - E_{k+1} \geq \frac{1}{12}h_k^3 g_k^2 + h_k^5 = h_k^3\left(\frac{1}{12}g_k^2 + h_k^2\right). \tag{K5}
$$

Thus E_k is strictly decreasing on the tail and is bounded below. Write $E_k \downarrow \ell$. Since $h_k \to 0$,

$$
f(s_k) \longrightarrow \ell, \qquad \sum_k h_k^3 g_k^2 < \infty. \tag{K6}
$$

No monotonicity of $f(s_k)$ itself has been claimed or used.

6.2 Criticality before finite length

Compactness gives $M := \sup_S \|\operatorname{grad} f\| < \infty$. Since $h_k \leq 1$, (K1) gives

$$
d_S(s_k, s_{k+1}) \leq (M/6 + R)h_k^3.
$$

The Hessian bound, with parallel transport along a short minimizing geodesic, implies

$$
|g_{k+1} - g_k| \leq L d_S(s_k, s_{k+1}) \leq D h_k^3 \tag{K7}
$$

for a fixed $D > 0$; increasing D to be at least 1 is harmless.

Here is a complete discrete slow-variation argument. If $g_k \not\to 0$, choose $\varepsilon > 0$ for which $g_k \geq \varepsilon$ infinitely often. Given such an index k, let $j > k$ be the first index satisfying

$$
\sum_{i=k}^{j-1} h_i^3 \geq \frac{\varepsilon}{2D}.
$$

The index exists because $\sum_i h_i^3 = \sum_i i^{-1} = \infty$. For every $k \leq i < j$, minimality of j and (K7) give $g_i \geq \varepsilon/2$. Therefore

$$
\sum_{i=k}^{j-1} h_i^3 g_i^2 \geq \frac{\varepsilon^2}{4}\sum_{i=k}^{j-1} h_i^3 \geq \frac{\varepsilon^3}{8D}.
$$

Infinitely many indices with $g_k \geq \varepsilon$ allow the construction of infinitely many disjoint such intervals. This contradicts (K6). Hence

$$g_k \to 0. \tag{K8}$$

Let Ω be the cluster set. It is nonempty and compact, $\operatorname{dist}(s_k, \Omega) \to 0$, and (K6), (K8) imply

$$\Omega \subset \{s : \operatorname{grad} f(s) = 0,\ f(s) = \ell\}. \tag{K9}$$

The distance-to-cluster-set assertion follows directly from compactness: any subsequence staying a fixed positive distance away would have a cluster point outside Ω.

6.3 Augmented gradient and time-scaled relative error

On $S \times \mathbb{R}$, with its product Riemannian metric, set

$$\mathcal{F}(s, z) = f(s) + Az^2, \qquad q_k = (s_k, h_k), \qquad G_k = \|\operatorname{grad} \mathcal{F}(q_k)\| = \sqrt{g_k^2 + 4A^2 h_k^2}.$$

Let $\rho_k = d_{S\times\mathbb{R}}(q_k, q_{k+1})$. From (K5),

$$E_k - E_{k+1} \geq c_0 h_k^3 G_k^2, \qquad c_0 := \min\{1/12,\ 1/(4A^2)\} > 0. \tag{K10}$$

Moreover, using $G_k \geq g_k$, $G_k \geq 2Ah_k$, and (K4),

$$\begin{aligned} d_S(s_k, s_{k+1}) &\leq h_k^3 \left(\frac{1}{6} g_k + R h_k\right) \leq B_s h_k^3 G_k, & B_s &:= \frac{1}{6} + \frac{R}{2A}, \\ \rho_k &\leq d_S(s_k, s_{k+1}) + (h_k - h_{k+1}) \leq B h_k^3 G_k, & B &:= B_s + \frac{1}{6A}. \end{aligned} \tag{K11}$$

For completeness there is also a reverse, time-scaled relative-error bound. The recurrence and the lower bound in (K4) imply

$$\begin{aligned} G_k &\leq g_k + 2Ah_k \\ &\leq \frac{6 d_S(s_k, s_{k+1})}{h_k^3} + (6R + 2A) h_k \\ &\leq \frac{6 d_S(s_k, s_{k+1}) + (6R + 2A)(h_k - h_{k+1})/\gamma_0}{h_k^3} \leq C_{\mathrm{rel}} \frac{\rho_k}{h_k^3}, \end{aligned} \tag{K12}$$

where $C_{\mathrm{rel}} = \sqrt{2}\max\{6, (6R+2A)/\gamma_0\}$ is sufficient. The Hessian norm of $\mathcal{F}$ is bounded by $L_{\mathcal{F}} := \max\{L, 2A\}$, so also

$$G_{k+1} \leq (C_{\mathrm{rel}} + L_{\mathcal{F}}) \frac{\rho_k}{h_k^3}.$$

Equations (K10)–(K11) imply the corresponding sufficient-decrease formulation

$$E_k - E_{k+1} \geq \frac{c_0}{B^2}\frac{\rho_k^2}{h_k^3}. \tag{K13}$$

The h_k^{-3} scaling is essential: this is a diminishing-step scheme. A uniform unscaled relative-error constant is neither needed nor asserted. Center-only steps can cancel against r_k, so (K12) is deliberately stated for the augmented product step. The finite-length proof below uses the particularly simple current-gradient bounds (K10)–(K11).

6.4 Power KL transfer and the length estimate

Apply (K2) at points of Ω. By (K9), all their critical values are the same ℓ. Choose a finite neighborhood cover of Ω; take the minimum of its positive constants μ_p, shrink the energy window to lie in $(0,1)$, and take the maximum of its finitely many exponents. Thus there are $\mu > 0$, $\eta \in (0,1]$, $\theta \in [0,1)$, and an open neighborhood $U \supset \Omega$ such that

$$g(s) \geq \mu(f(s) - \ell)^\theta \quad \text{for } s \in U, \quad 0 < f(s) - \ell < \eta. \tag{K14}$$

The exponent replacement is valid because $t^\theta \leq t^{\theta_p}$ for $0 < t \leq 1$ and $\theta \geq \theta_p$. Eventually $s_k \in U$ and $|f(s_k) - \ell| < \eta$.

Put

$$\vartheta := \max\{\theta, 1/2\} < 1, \qquad e_k := E_k - \ell.$$

Strict decrease in (K5) implies $e_k > 0$ at every finite index on the tail. We claim that, eventually,

$$e_k^\vartheta \leq C_{\mathrm{KL}} G_k, \qquad C_{\mathrm{KL}} := \mu^{-1} + \tfrac{1}{2}A^{\vartheta-1}. \tag{K15}$$

Indeed, let $t_k = f(s_k) - \ell$. If $t_k > 0$, subadditivity of $x \mapsto x^\vartheta$, (K14), $t_k \leq 1$, and $2\vartheta \geq 1$ give

$$\begin{aligned} e_k^\vartheta = (t_k + Ah_k^2)^\vartheta &\leq t_k^\vartheta + A^\vartheta h_k^{2\vartheta} \\ &\leq \mu^{-1} g_k + A^\vartheta h_k \leq C_{\mathrm{KL}} G_k. \end{aligned}$$

If $t_k \leq 0$, then $0 < e_k \leq Ah_k^2$, so the same conclusion follows using only the last quadratic term. Thus the argument requires only the one-sided KL inequality (K14), even though $f(s_k)$ need not approach ℓ from above.

Define the increasing concave desingularizer

$$\Phi(e) = \frac{C_{\mathrm{KL}}}{1-\vartheta} e^{1-\vartheta}, \qquad e > 0, \qquad \Phi(0) = 0.$$

Equation (K15) says $\Phi'(e_k)G_k \geq 1$. Concavity, (K10), and then (K11) give

$$\begin{aligned}\Phi(e_k)-\Phi(e_{k+1}) &\ge \Phi'(e_k)(e_k-e_{k+1})\\ &\ge c_0 h_k^3 \Phi'(e_k) G_k^2\\ &\ge c_0 h_k^3 G_k \ge \frac{c_0}{B_s} d_S(s_k,s_{k+1}).\end{aligned} \tag{K16}$$

Summing and letting the terminal index tend to infinity proves

$$\sum_{k=K}^{\infty} d_S(s_k,s_{k+1}) \le \frac{B_s C_{\mathrm{KL}}}{c_0(1-\vartheta)}(E_K-\ell)^{1-\vartheta} < \infty \tag{K17}$$

for every sufficiently large K. Replacing B_s by B proves finite length of the product sequence (s_k,h_k) as well. Finite center length makes (s_k) Cauchy; compactness gives its limit, and (K8) makes that limit critical. This proves the power-KL assertion.

6.5 General concave KL version

The power assumption can be replaced by the following usual concave KL form at every potential critical cluster point p: there are U_p, $\eta_p > 0$, and a continuous function $\varphi_p : [0,\eta_p) \to [0,\infty)$, with $\varphi_p(0) = 0$, such that φ_p is concave, C^1 on $(0,\eta_p)$, has $\varphi_p' > 0$, and

$$\varphi_p'(f(s)-f(p))\,\|\operatorname{grad} f(s)\| \ge 1 \quad \text{when } s \in U_p,\ 0 < f(s)-f(p) < \eta_p. \tag{K18}$$

The augmented-energy, criticality, and relative-error arguments above do not use KL. Once their common-value compact cluster set Ω is known, take a finite cover as before and set $\varphi = \sum_{i=1}^{N} \varphi_{p_i}$, restricting every term to a common energy interval. Then φ remains a concave desingularizer and satisfies the uniform version of (K18) near Ω, with critical value ℓ.

For sufficiently small $e > 0$, define

$$\widehat{\Phi}(e) := 2\varphi(e/2) + \sqrt{2/A}\,\sqrt{e}, \qquad \widehat{\Phi}'(e) = \varphi'(e/2) + \frac{1}{\sqrt{2Ae}}. \tag{K19}$$

This is continuous at zero, zero there, increasing, concave, and C^1 for $e > 0$. To check the augmented KL inequality, write

$$e = f(s) - \ell + Az^2 > 0, \qquad G = \sqrt{\|\operatorname{grad} f(s)\|^2 + 4A^2 z^2}.$$

If $f(s)-\ell \ge e/2$, then $0 < f(s)-\ell \le e$. Concavity makes φ' nonincreasing, so

$$\varphi'(e/2)G \ge \varphi'(f(s)-\ell)\|\operatorname{grad} f(s)\| \ge 1.$$

Otherwise $Az^2 > e/2$, and $G \ge 2A|z| > \sqrt{2Ae}$. In either case,

$$\widehat{\Phi}'(e)G \geq 1. \tag{K20}$$

The telescoping proof (K16) now applies verbatim with $\widehat{\Phi}$, and yields

$$\sum_{k=K}^{\infty} d_S(s_k, s_{k+1}) \leq \frac{B_s}{c_0}\widehat{\Phi}(E_K - \ell) < \infty. \tag{K21}$$

Thus **the general concave local KL property suffices**; no power exponent or analyticity is essential to this deterministic lemma.

The same telescoping inequalities give $\sum_k h_k^3 G_k < \infty$, and hence $\sum_k h_k^3 \| \operatorname{grad} f(s_k)\| < \infty$ in either case. Finite length makes the center sequence Cauchy, and compactness gives its limit; the already proved gradient convergence makes the limit critical. ∎

7 Secondary criticality and finite actual paths

Set $f = -\psi|_S$. The C^5/C^5 geometry makes H at least C^3 as a normal-bundle field, ν at least C^4, and ψ at least C^3. The selected critical submanifold S is C^4 with its induced C^3 metric. Thus it has precisely the finite regularity required in Lemma 3. No infinite differentiability is being inferred.

Equation (76) is the actual recurrence

$$\operatorname{Log}_{s_k}(s_{k+1}) = -\frac{h_k^3}{6}\operatorname{grad}_S f(s_k) + r_k, \qquad |r_k| \leq R h_k^4.$$

The energy and criticality parts of Lemma 3, which use no KL assumption, therefore show

$$f(s_k) \to \ell, \qquad \operatorname{grad}_S f(s_k) \to 0.$$

Since $n_k \to 0$ and $p_k - s_k \to 0$, the actual common cluster set belongs to $\operatorname{Crit}_S \psi$. This completes part B, still before one-point convergence.

Under the additional power or general concave KL hypothesis, the lemma proves finite center length and $s_k \to p_* \in \operatorname{Crit}_S \psi$. It also gives the stronger summability

$$\sum_k h_k^3 |\operatorname{grad}_S \psi(s_k)| < \infty. \tag{78}$$

Use (77), $W = \operatorname{grad}_S \psi/6$, and $\sum h_k^4 < \infty$ to obtain summability of the actual histories. Index shifting changes only a finite prefix, so

$$\sum_{k \geq K} |x_{k+1} - x_k| < \infty.$$

Vanishing normal and primary transverse offsets give $x_k \to p_*$. The literal predictor offset obeys $|y_k - x_k| \leq C|x_k - x_{k-1}|$, by bounded Hessian of F on the controlled neighborhood. Thus $y_k \to p_*$, the offsets are summable, and

$$|y_{k+1} - y_k| \leq |x_{k+1} - x_k| + |y_{k+1} - x_{k+1}| + |y_k - x_k|$$

proves finite length of the actual predictor path. This proves part C and Theorem 1.

The order-h^4 tangent error in the lemma is quantitative and uniform on a late suffix. Merely knowing that an unspecified perturbation is summable is not its hypothesis. Likewise, the order-h^4 normal-history estimate was proved before actual path summation. Center finite length alone was not used as a substitute for that history control.

7.1 Why secondary regularity is an additional assumption

The secondary KL assumption cannot be removed by declaring the primary geometry analytic. For example, on the unit-circle annulus let

$$H(\varphi) = 2 + \tfrac{1}{10}\chi(\varphi), \qquad F(r,\varphi) = \tfrac{1}{2}H(\varphi)(r-1)^2,$$

where $\chi(\varphi) = \exp(-1/\sin^2(\varphi/2))$ away from integer multiples of 2π, and $\chi = 0$ at those multiples. The function is smooth and periodic, flat but nonanalytic at 0. Here $\psi = 1/(2H) > 0$. Analyticity of ψ at 0 would imply analyticity of H, which is false. Moreover, $f = -\psi$ has a flat strict local minimum at 0, and for every $\theta < 1$,

$$\frac{|f'(\varphi)|}{(f(\varphi) - f(0))^\theta} \longrightarrow 0 \quad (\varphi \to 0).$$

Indeed differentiation introduces only polynomial powers of φ^{-1}, whereas the numerator after division retains a positive exponential factor of the form $\chi^{1-\theta}$. Thus even a power KL inequality is not automatic for arbitrary smooth secondary data. The general concave alternative is a separate, broader hypothesis.

The finite-cover uniformization used in the lemma is the intrinsic counterpart of the compact constant-level construction in the author manuscript [4, Lemma 3.6, pp. 20–21]; it is proved here directly. The general concave formulation belongs to the gradient-inequality framework of [5]. No external diminishing-step convergence theorem is invoked, and no analyticity or definability of the present f is silently assumed.

The capture proof uses a cumulative geometric-transient velocity barrier and a summable $O(h^5)$ primary energy remainder. It never uses the divergent $\sum h_k^3$ as an error budget. Its geometry, normal metric, moving-bundle analysis and deterministic KL argument are all established above. Finite metric regularity, the current-base history, and the signed unaugmented-level split are retained throughout.

The general theorem asserts critical selection, not a global maximum of ψ, a globally least-norm point, or an arbitrary-start basin. Its center-tail estimate is the augmented-energy bound of Lemma 3. The following model has additional restoring structure and consequently permits stronger, entry-uniform basin conclusions and an explicit fixed-entry rate.

8 Uniform stiffness selection on a circle

On the annulus

$$\mathcal{A} = \{x \in \mathbb{R}^2 : 3/4 < |x| < 5/4\},$$

write $x = re(\varphi)$, where

$$e(\varphi) = (\cos\varphi, \sin\varphi), \qquad t(\varphi) = (-\sin\varphi, \cos\varphi),$$

and define

$$F(r, \varphi) = \tfrac{1}{2}H(\varphi)(r-1)^2, \qquad H(\varphi) = 2 + \tfrac{1}{2}\sin\varphi. \tag{79}$$

The formula is well defined and smooth on $\mathcal{A}$. Its minimum set there is the whole unit circle. The restricted squared norm $q(x) = |x|^2/2$ is constant on that circle. The normal Hessian is $H(\varphi) \in [3/2, 5/2]$; it has its unique minimum at

$$p_{\min} = (0, -1).$$

Thus minimum stiffness distinguishes a point even though neither the objective value nor the norm distinguishes the circle's minimizers.

We use precisely the clock (3) and literal map (4). Since $q|_M$ is constant, $S = M$, V_s has rank zero, and $\psi(\varphi) = 1/(2H(\varphi))$. The next statements strengthen the general theorem with constants uniform from the original entry and a specified basin. In this section only, n_k, u_k denote scalar radial coordinates and w_k denotes the unscaled tangential history; thus $w_k = h_k v_k$ in scalar tangent coordinates.

At an actual iterate define the current-frame, unscaled history variables by

$$\begin{aligned} x_k &= (1 + n_k)e(\varphi_k), \\ x_k - x_{k-1} &= u_k e(\varphi_k) + w_k t(\varphi_k). \end{aligned} \tag{80}$$

Here w_k is the actual tangential history component, with no factor of h_k removed. The small-increment lift φ_k will be justified by capture.

Theorem 2 (Critical-clock stiffness selection)**.** *There is a finite integer K_0, independent of the entry integer K, such that for every $K \geq K_0$, the initialization*

$$x_{K-1} = x_K = (1, 0) \tag{81}$$

has the following properties. All actual iterates and predictors stay in $\mathcal{A}$, and

$$x_k \longrightarrow p_{\min}, \qquad y_k \longrightarrow p_{\min}. \tag{82}$$

Both actual paths have finite length:

$$\sum_{k \geq K} |x_{k+1} - x_k| < \infty, \qquad \sum_{k \geq K} |y_{k+1} - y_k| < \infty. \tag{83}$$

More generally the same assertions hold for the entry family

$$\varphi_K \in [-\pi, 0], \qquad |n_K| + |u_K| \le h_K^2, \qquad |w_K| \le h_K^3, \tag{84}$$

with x_K, x_{K-1} reconstructed by (80). The lifted angle then remains in $|\varphi_k + \pi/2| < 3\pi/4$.

Put $\psi(\varphi) = 1/(2H(\varphi))$. There are $C, c > 0$ and $\rho \in (0,1)$, uniform over this entry family and all $K \ge K_0$, such that, with

$$T_k = h_K^2 \rho^{k-K}, \qquad E_{k,K} = \exp\left(-c \sum_{j=K}^{k-1} h_j\right),$$

one has

$$n_k = -\frac{h_k^2}{H(\varphi_k)} + O(h_k^4 + T_k), \qquad u_k = O(h_k^4 + T_k), \tag{85}$$

$$w_k = \frac{h_k^3}{6}\psi'(\varphi_k) + O(h_k^4 + h_K^3 E_{k,K}), \tag{86}$$

$$\varphi_{k+1} - \varphi_k = \frac{\psi'(\varphi_k)}{6k} + r_k, \qquad |r_k| \le C(h_k^4 + h_K^3 E_{k,K}), \tag{87}$$

$$\sum_{k \ge K} |r_k| \le C h_K = C K^{-1/3}. \tag{88}$$

All constants in these estimates are independent of K.

Theorem 3 (Rates after selection)**.** *For each fixed admitted K and history in Theorem 2,*

$$|\varphi_k + \pi/2| = O(k^{-1/54}), \qquad \lim_{k\to\infty} k^{1/54}(\varphi_k + \pi/2) \in \mathbb{R}, \tag{89}$$

and

$$\sum_{j \ge k} \big(|x_{j+1} - x_j| + |y_{j+1} - y_j|\big) = O(k^{-1/54}). \tag{90}$$

The scaled limit may be zero. The constants in this theorem may depend on the fixed entry time and history.

The estimates below are proved directly from the same literal map to retain uniformity in the original entry. Since $h_k^3 = 1/k$ is not summable, the raw bound $|\varphi_{k+1} - \varphi_k| = O(h_k^3)$ does not prove finite travel. The restoring argument in Section 8.5 is essential. The domain assumption is annular and the entry is sufficiently late on the original clock; no global or arbitrary-early capture statement is made.

8.1 The exact raw map and local estimates

Throughout the proof constants are finite and uniform in φ and in the original entry time. They may be enlarged in the order indicated, but not in response to a chosen K or an unknown limiting point. At a current actual state abbreviate

$$x = (1+n)e, \qquad d = x - x_{\text{prev}} = ue + wt, \qquad z = (n,u), \quad S = |z|.$$

The predecessor is exactly $(1+n-u)e - wt$. For a nearby point $(1+s)e + vt$ set

$$R = \sqrt{(1+s)^2 + v^2}, \qquad \alpha = \arctan \frac{v}{1+s}.$$

The gradient in the current frame is exactly

$$\begin{aligned} G_e(s,v) &= H(\varphi+\alpha)(R-1)\cos\alpha - \frac{H'(\varphi+\alpha)(R-1)^2 \sin\alpha}{2R}, \\ G_t(s,v) &= H(\varphi+\alpha)(R-1)\sin\alpha + \frac{H'(\varphi+\alpha)(R-1)^2 \cos\alpha}{2R}. \end{aligned} \tag{91}$$

Choose a fixed small compact coordinate neighborhood, and a slightly larger one for the raw map, both contained in $\mathcal{A}$. Smoothness and periodicity give uniform Taylor estimates

$$\begin{aligned} G_e(s,v) &= Hs + O(|s||v| + v^2), \\ G_t(s,v) &= \tfrac{1}{2}H's^2 + O(|s|^3 + |s||v| + v^2). \end{aligned} \tag{92}$$

In fact $G_e(s,0) = Hs$ and $G_t(s,0) = H's^2/[2(1+s)]$ exactly. The latter has s derivative $O(|s|)$. For the actual two-point difference $D = \nabla F(x) - \nabla F(x_{\text{prev}})$ this yields

$$D_e = Hu + O(S|w| + w^2), \qquad |D_t| \le C(S|u| + S|w| + w^2). \tag{93}$$

For the second estimate, compare $G_t(n,0)$ first with $G_t(n-u,0)$ and then with $G_t(n-u,-w)$. Mean-value estimates on these two segments give the three terms displayed. This is a difference of actual gradients, not a differentiated eventual asymptotic formula.

Write $A = 1 - bH$, so $3/8 \le A \le 5/8$. If $y = (1+s_y)e + v_y t$, the literal predictor gives

$$s_y = n + (A - \gamma h)u + O(S|w| + w^2), \qquad v_y = (1-\gamma h)w - bD_t. \tag{94}$$

Writing $x_+ = (1+s_+)e + v_+^{\text{old}} t$, the correction is

$$\begin{aligned} s_+ &= s_y - bG_e(s_y, v_y) - bh^2(1+s_y), \\ v_+^{\text{old}} &= v_y - bG_t(s_y, v_y) - bh^2 v_y. \end{aligned} \tag{95}$$

For $S + |w| + h^2$ sufficiently small, all points lie in the prescribed larger neighborhood and have positive radial coordinate. Let $\delta = \arg(x_+) - \varphi$ be the unique small lift, $n_+ = |x_+| - 1$, and let u_+, w_+ be the components of $x_+ - x$ in the new frame. Exact geometry gives

$$\begin{aligned} n_+ &= s_+ + O((v_+^{\text{old}})^2), \\ u_+ &= (n_+ - n) + (1+n)(1-\cos\delta), \\ w_+ &= (1+n)\sin\delta = \frac{1+n}{1+n_+} v_+^{\text{old}}. \end{aligned} \tag{96}$$

Lemma 4 (Uniform one-step bounds before capture)**.** *On the fixed small neighborhood above,*

$$z_+ = B(\varphi)z - bh^2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \gamma h A u \begin{pmatrix} 1 \\ 1 \end{pmatrix} + R_N, \quad B(\varphi) = \begin{pmatrix} A & A^2 \\ A-1 & A^2 \end{pmatrix}, \tag{97}$$

$$|R_N| \le C(h^2 S + S|w| + w^2 + S^3), \tag{98}$$

$$|w_+ - (1-\gamma h)w| \le C\big[(h^2+S)|w| + w^2 + S^2\big], \tag{99}$$

$$|\delta| \le C(|w| + S^2). \tag{100}$$

Proof. Before imposing capture caps, (94) gives

$$|s_y| \le C(S + w^2), \qquad |v_y| \le C(|w| + S^2).$$

The radial linear part of (95) is $A[n+(A-\gamma h)u]-bh^2$. The remaining terms are controlled by $h^2|s_y|$, $|s_y v_y|$, v_y^2, and $S|w| + w^2$. For example,

$$|s_y v_y| \le C(S|w| + S^3 + |w|^3 + S^2 w^2).$$

On a fixed small neighborhood, the extra products are absorbed by $C(w^2 + S^3)$; also $v_y^2 \le C(w^2+S^4)$ and $h^2 w^2 \le Cw^2$. The first two identities of (96) add at most $C(|w|+S^2)^2$. This proves (97)–(98).

For the tangent estimate use (93) and

$$|G_t(s_y, v_y)| \le C(S^2 + S|w| + w^2).$$

The ratio $(1+n)/(1+n_+)$ is $1 + O(S + h^2 + w^2)$, while $v_+^{\mathrm{old}} = O(|w| + S^2)$. Its additional products are all absorbed by the right side of (99). Finally, the arctangent expression for δ has a denominator bounded away from zero and gives (100). These are estimates for the actual map before any infinite-time bound is assumed. ∎

8.2 Normal contraction and annular capture

For this stiffness interval the Euclidean norm suffices. The Frobenius bound gives

$$\|B(\varphi)\|^2 \le A^2 + (A-1)^2 + 2A^4 \le \frac{17}{32} + \frac{625}{2048} = \frac{1713}{2048} < \left(\frac{15}{16}\right)^2. \tag{101}$$

Set $\rho_0 = 15/16$, $\rho = 31/32$ and $D_0 = 64$. Use a single $C \ge 1$ for Lemma 4 and choose $M \ge 1$ so large that

$$CD_0^2 \le \gamma M/4. \tag{102}$$

Now choose $h_* > 0$ small enough that the neighborhood conditions hold under

$$S \le D_0 h^2, \qquad |w| \le Mh^3, \qquad 0 < h \le h_*, \tag{103}$$

and so that, for every such h,

$$\gamma \tfrac{5}{8}\sqrt{2}\, h + C(h^2 + Mh^3 + D_0^2 h^4) \le \rho - \rho_0, \tag{104}$$

$$b\sqrt{2} + CM^2 h^4 \le 1, \tag{105}$$

$$C[(1+D_0)h^2 + Mh^3] \le \gamma h/4, \tag{106}$$

$$\tfrac{2}{3}h^3 \le \tfrac{1}{128}, \qquad h^2 \le \gamma/4, \qquad \gamma h \le \tfrac{1}{2}. \tag{107}$$

The constants D_0, M are fixed first. All these inequalities are compatible as $h \downarrow 0$; after dividing (106) by h, its left side tends to zero. The entry bounds (84) imply the initial caps.

Under (103), the normal estimates give

$$|z_+| \le \rho|z| + h^2 \le 63h^2. \tag{108}$$

The original clock has the exact relation

$$\frac{h_{k+1}^2}{h_k^2} = (1+h_k^3)^{-2/3} \ge 1 - \tfrac{2}{3}h_k^3 \ge \tfrac{127}{128}. \tag{109}$$

Thus $63h_k^2 < 64h_{k+1}^2$. Likewise,

$$\begin{aligned} |w_+| &\le (1-3\gamma h/4)Mh^3 + CD_0^2h^4 \\ &\le Mh^3(1-\gamma h/2) < Mh_{k+1}^3, \end{aligned}$$

because $h_{k+1}^3/h_k^3 = (1+h_k^3)^{-1} \ge 1 - h_k^3$ and $h_k^2 \le \gamma/4$. These are strict improvements of both caps.

A finite-step induction, applied in the larger coordinate neighborhood at a possible first exit, proves

$$|n_k| + |u_k| \le \sqrt{2}D_0h_k^2, \qquad |w_k| \le Mh_k^3, \qquad |\delta_k| \le Ch_k^3 \tag{110}$$

for all $k \ge K$ whenever $h_K \le h_*$. The predecessor, predictor and next iterate stay in $\mathcal{A}$ at every step. Decrease h_* once more if necessary to put their radii in $[7/8, 9/8]$. In particular, the small-increment angle lift is defined for the entire actual suffix. No supremum over an unproved infinite orbit is used. This establishes annular capture for the new schedule.

8.3 Actual normal tracking with the entry transient

Center only the actual recurrence already established. Define the algebraic reference

$$a_k = \left(-\frac{h_k^2}{H(\varphi_k)}, 0\right), \qquad \zeta_k = z_k - a_k.$$

It obeys the exact identity

$$B(\varphi_k)a_k - bh_k^2\begin{pmatrix}1\\1\end{pmatrix} = a_k. \tag{111}$$

The damping matrix in (97) vanishes on a_k, and (104) gives

$$\left\| B(\varphi_k) - \gamma h_k \begin{pmatrix} 0 & A \\ 0 & A \end{pmatrix} \right\| \le \rho.$$

Under capture, $|R_N| \le Ch_k^4$. Moreover the original clock and (110) give

$$|h_{k+1}^2 - h_k^2| \le \tfrac{2}{3}h_k^5, \qquad |H(\varphi_{k+1})^{-1} - H(\varphi_k)^{-1}| \le C|\delta_k| \le Ch_k^3.$$

Consequently $|a_{k+1} - a_k| \le Ch_k^5$ and

$$|\zeta_{k+1}| \le \rho|\zeta_k| + Ch_k^4. \tag{112}$$

Since $|\zeta_K| \le Ch_K^2$, iteration proves

$$|\zeta_k| \le C(h_k^4 + h_K^2\rho^{k-K}). \tag{113}$$

For completeness, if $\ell = k - 1 - j$ and $j \ge 1$, then

$$\frac{h_j^4}{h_k^4} = (k/j)^{4/3} \le (\ell+2)^{4/3}, \qquad \sum_{\ell\ge 0} \rho^\ell(\ell+2)^{4/3} < \infty.$$

This directly bounds the geometric forcing convolution uniformly in K. The same polynomial comparison with exponent $2/3$ gives $T_k = h_K^2\rho^{k-K} \le Ch_k^2$. We have proved

$$n_k = -\frac{h_k^2}{H(\varphi_k)} + O(h_k^4 + T_k), \qquad u_k = O(h_k^4 + T_k). \tag{114}$$

In particular $\sum_{k\ge K}|u_k| < \infty$, uniformly bounded for sufficiently late entries. The entry transient has not been suppressed. The reference a_k centers a proven actual recurrence; it is not a stationary branch substituted for the iterates.

8.4 Raw angular forcing and actual velocity filtering

8.4.1 The forcing coefficient from both algorithm lines

Fix a current original index. From (93), capture and (114),

$$D_t = O(h^5 + h^2T_k). \tag{115}$$

Indeed $S|u| = O(h^6 + h^2T_k)$, $S|w| = O(h^5)$ and $w^2 = O(h^6)$. Therefore the actual predictor satisfies

$$\begin{aligned} s_y &= -h^2/H(\varphi) + O(h^4 + T_k), \\ v_y &= (1 - \gamma h)w + O(h^5 + h^2T_k). \end{aligned} \tag{116}$$

In particular $s_y = O(h^2)$ and $v_y = O(h^3)$. Evaluating the actual gradient at this actual predictor gives

$$G_t(s_y, v_y) = \frac{H'(\varphi)}{2H(\varphi)^2}h^4 + O(h^5 + h^2T_k). \tag{117}$$

To check the error, the difference of s_y^2 from h^4/H^2 is bounded by $Ch^2(h^4 + T_k)$, using $T_k \le Ch^2$. The Taylor remainders in (92) have orders h^6, h^5, h^6. The regularization contributes $h^2v_y = O(h^5)$. Thus the literal old-frame correction is

$$v_+^{\text{old}} = (1 - \gamma h)w - \frac{bH'}{2H^2}h^4 + O(h^5 + h^2T_k).$$

By capture, $(1+n)/(1+n_+) = 1 + O(h^2)$ and $v_+^{\mathrm{old}} = O(h^3)$. With $\psi = 1/(2H)$ and $\psi' = -H'/(2H^2)$, the next-frame actual history obeys

$$\begin{aligned} w_{k+1} &= (1-\gamma h_k) w_k + b h_k^4 \psi'(\varphi_k) + R_k, \\ |R_k| &\le C(h_k^5 + h_k^2 T_k). \end{aligned} \tag{118}$$

This calculation retains both the unregularized gradient-history term and regularization at y_k; their additional angular contributions are in the remainder. Finally, the exact identity for w_+ in (96), followed by the small-argument arcsine formula, gives

$$\delta_k = w_{k+1} + O(h_k^5). \tag{119}$$

Here the radial denominator changes w_{k+1} by $O(h_k^5)$, and the cubic arcsine term is even smaller.

8.4.2 The slow-clock kernel

Set

$$f = \frac{b}{\gamma}\psi' = \frac{1}{6}\psi', \qquad \eta_k = w_k - h_k^3 f(\varphi_k).$$

This is an actual velocity defect. Since $|\delta_k| = O(h_k^3)$, f is uniformly Lipschitz and

$$|h_{k+1}^3 - h_k^3| = \frac{h_k^6}{1+h_k^3} \le h_k^6,$$

one has

$$|h_{k+1}^3 f(\varphi_{k+1}) - h_k^3 f(\varphi_k)| \le C h_k^6.$$

The identity $\gamma f = b\psi'$ cancels the leading terms of (118), giving

$$\eta_{k+1} = (1-\gamma h_k)\eta_k + q_k, \qquad |q_k| \le C(h_k^5 + h_k^2 T_k), \quad |\eta_K| \le C h_K^3. \tag{120}$$

The damping factors lie in $(0,1)$ by (107). Define

$$t_{k,j} = \sum_{m=j}^{k-1} h_m, \qquad P_{k,j} = \prod_{m=j}^{k-1} (1-\gamma h_m).$$

Then $P_{k,j} \le e^{-\gamma t_{k,j}}$. Also, for $K \le j < k$,

$$\log(k/j) \le \sum_{m=j}^{k-1} \frac{1}{m} \le h_K^2 t_{k,j}. \tag{121}$$

Decrease the already uniform h_* so that

$$\tfrac{4}{3} h_*^2 \le \gamma/2, \qquad \rho e^{\gamma h_*} < 1. \tag{122}$$

The ratio h_j^4/h_k^4 then uses at most half of the damping exponent. Accordingly,

$$\sum_{j=K}^{k-1} e^{-\gamma t_{k,j+1}} h_j^5 \le C h_k^4. \tag{123}$$

To see this without asymptotic shorthand, factor out h_k^4 and use (121); the remaining sum is at most a fixed multiple of

$$\sum_{j=K}^{k-1} h_j e^{-(\gamma/2) t_{k,j+1}}.$$

Compare on the adjacent slow-time intervals of length h_j. It is bounded by the integral of a decaying exponential times a fixed factor $e^{\gamma h_*}$.

The normal transient can be bounded separately using $h_j \le h_K$ and $t_{j+1,K} \le (j - K + 1)h_K$:

$$\sum_{j=K}^{k-1} e^{-\gamma t_{k,j+1}} h_j^2 h_K^2 \rho^{j-K} \le C h_K^4 e^{-\gamma t_{k,K}}. \tag{124}$$

After factoring out $h_K^4 e^{-\gamma t_{k,K}}$, the residual geometric sum has ratio at most $\rho e^{\gamma h_*} < 1$. Variation of constants in (120) therefore yields

$$|\eta_k| \le C h_k^4 + C h_K^3 e^{-\gamma t_{k,K}}. \tag{125}$$

This includes both the actual initial velocity defect and the effect of the normal entry transient.

For the position equation, combine (118), (119) and (125). The leading damping loss $\gamma h_k \cdot h_k^3 f$ cancels $b h_k^4 \psi'$ exactly. We obtain

$$\delta_k = \frac{f(\varphi_k)}{k} + r_k, \qquad |r_k| \le C\left(h_k^4 + h_K^3 e^{-c t_{k,K}}\right) \tag{126}$$

for a fixed $c > 0$, for example $c = \gamma/2$ after enlarging C. The term $h_k^2 T_k$ is absorbed by the slower entry exponential under (122). This proves the actual harmonic angle equation, not just a formal substitution into a stationary normal displacement.

8.4.3 A uniform small total error budget

The power term satisfies

$$\sum_{k \ge K} h_k^4 = \sum_{k \ge K} k^{-4/3} \le C h_K.$$

For the exponential term, (121) gives

$$\frac{h_K}{h_k} = (k/K)^{1/3} \le e^{(h_K^2/3) t_{k,K}}.$$

Decrease h_* further so that $h_*^2/3 \le c/2$. Insert a factor h_k and compare the forward slow-time intervals to the exponential integral. This proves

$$\sum_{k\ge K} e^{-ct_{k,K}} \le C/h_K. \tag{127}$$

For example, $h_k e^{-(c/2)t_{k,K}}$ is bounded by a fixed multiple of the integral of $e^{-(c/2)t}$ over $[t_{k,K}, t_{k+1,K}]$, so its sum is uniformly bounded. Hence

$$\sum_{k\ge K} |r_k| \le C_R(h_K + h_K^2) \le 2C_R h_K, \tag{128}$$

after taking $h_* \le 1$. All constants are uniform over the admitted entry family and over sufficiently late K. The velocity estimate (86), the angle estimate (87) and their uniform budget are now established before any selection claim.

8.5 Basin confinement and restoring selection

Put $\theta_k = \varphi_k + \pi/2$. Then

$$H(\varphi) = 2 - \tfrac{1}{2}\cos\theta, \qquad f(\varphi) = -\frac{\sin\theta}{24(2-\tfrac{1}{2}\cos\theta)^2}. \tag{129}$$

On $0 < |\theta| < \pi$ the drift has sign $-\operatorname{sign}\theta$, and its only zero in $(-\pi,\pi)$ is $\theta = 0$. Let $L \ge 1$ be a global Lipschitz constant for f. Fix the final entry threshold to satisfy

$$K_0^{-1/3} \le h_*, \qquad K_0 \ge L, \qquad 2C_R K_0^{-1/3} < \pi/8. \tag{130}$$

These are inequalities on previously fixed finite constants and specify a finite uniform late-start threshold. No value for the smallest possible K_0 is claimed.

As long as $|\theta_k| < 3\pi/4$, the unperturbed step $\theta_k + f(\varphi_k)/k$ cannot cross zero: its magnitude of motion is at most $L|\theta_k|/k \le |\theta_k|$. Thus (126) gives the decisive inequality

$$|\theta_{k+1}| \le |\theta_k| - \frac{|f(\varphi_k)|}{k} + |r_k|. \tag{131}$$

Every entry in (84) satisfies $|\theta_K| \le \pi/2$. Finite-prefix induction, using (128), gives

$$|\theta_k| \le \pi/2 + \sum_{j=K}^{k-1} |r_j| < 5\pi/8 < 3\pi/4. \tag{132}$$

This closes the angular basin without assuming any eventual limit. Summing (131) proves

$$\sum_{k\ge K} \frac{|f(\varphi_k)|}{k} \le |\theta_K| + \sum_{k\ge K} |r_k| < \infty. \tag{133}$$

Together with (126), it follows that

$$\sum_{k\ge K} |\varphi_{k+1} - \varphi_k| < \infty. \tag{134}$$

The angle therefore has a limit in the compact interior basin of (132). If f were nonzero at that limit, continuity and $\sum 1/k = \infty$ would contradict (133). Its only zero in this basin is $\varphi = -\pi/2$, so

$$\varphi_k \longrightarrow -\pi/2. \tag{135}$$

For the start $(1, 0)$ this changes stiffness from $H(0) = 2$ to $H(-\pi/2) = 3/2$. The cumulative leading angular time is harmonic and divergent; the finite total variation comes from the restoring sign, not from summability of $1/k$.

8.6 Finite length of both actual paths

The normal estimate (114) gives $\sum |u_k| < \infty$. From the exact second identity in (96) and $|\delta_k| = O(1/k)$,

$$\sum_{k \geq K} |n_{k+1} - n_k| \leq \sum_{k \geq K} |u_{k+1}| + C \sum_{k \geq K} \delta_k^2 < \infty. \tag{136}$$

As

$$|x_{k+1} - x_k| \leq |n_{k+1} - n_k| + (1 + |n_k|)|\delta_k|,$$

(134) and (136) prove finite length of the actual x sequence. Also $n_k \to 0$, so (135) proves $x_k \to (0, -1)$.

The gradient is uniformly Lipschitz on a compact neighborhood containing every short segment between x_k and x_{k-1}. The literal predictor gives

$$|y_k - x_k| \leq C|x_k - x_{k-1}|. \tag{137}$$

The right side is summable, including the finite entry term. Hence $\sum |y_k - x_k| < \infty$, $y_k \to (0, -1)$ and

$$\sum_{k \geq K} |y_{k+1} - y_k| \leq \sum_{k \geq K} |x_{k+1} - x_k| + 2 \sum_{k \geq K} |y_k - x_k| < \infty. \tag{138}$$

These are lengths of the literal sequences, not only of their circle projections or of a reference curve. This completes the proof of Theorem 2.

8.7 Fixed-entry rates after selection

Fix an admitted K and history. Since $t_{k,K}$ grows as a positive constant times $k^{2/3}$, the entry exponential in (126) is smaller than every power of k on this orbit. Therefore $r_k = O(k^{-4/3})$. The drift in (129) is odd in θ and has the expansion

$$f(-\pi/2 + \theta) = -a\theta + O(\theta^3), \qquad a = 1/54. \tag{139}$$

We use the already established $\theta_k \to 0$. For every $0 < \alpha < a$, the recurrence eventually implies

$$|\theta_{k+1}| \leq (1 - \alpha/k)|\theta_k| + Ck^{-4/3}.$$

Products of $1-\alpha/j$ are bounded by a constant times $(j/k)^{\alpha}$. Variation of constants and $\alpha < 1/3$ then give $\theta_k = O(k^{-\alpha})$.

Choose now $a/3 < \alpha < a$. Returning to the signed equation yields

$$\theta_{k+1} = (1-a/k)\theta_k + s_k, \qquad |s_k| \le C\big(k^{-1-3\alpha} + k^{-4/3}\big). \tag{140}$$

For a sufficiently late fixed j_0 define

$$Q_k = \prod_{j=j_0}^{k-1}(1-a/j) > 0.$$

Summing $\log(1-a/j) + a/j = O(j^{-2})$ shows that $k^a Q_k$ has a finite positive limit. Thus

$$\sum_{k\ge j_0} \frac{|s_k|}{Q_{k+1}} \le C \sum_{k\ge j_0} \big(k^{a-1-3\alpha} + k^{a-4/3}\big) < \infty.$$

Variation of constants in (140) proves that θ_k/Q_k has a finite signed limit. It follows that $k^a\theta_k$ has a finite limit and $\theta_k = O(k^{-a})$. This does not assert a nonzero coefficient.

Summing (131) from k onward and using (126) gives

$$\sum_{j\ge k} |\delta_j| \le |\theta_k| + 2\sum_{j\ge k} |r_j| = O(k^{-a}). \tag{141}$$

By (136), the normal-variation tail is $O(k^{-1/3})$ plus a geometric transient. Hence the actual x-path tail is $O(k^{-a})$. Inequality (137) gives the same result for the actual y-path tail. Its one-index offset is harmless because $|x_k - x_{k-1}|$ is bounded by the preceding x-tail. This proves Theorem 3. All rate constants in this section concern a fixed admitted entry.

8.8 Compact subarcs away from maximum stiffness

The normal and velocity estimates above use only periodic uniform bounds and the normal and history caps. This permits a larger angular basin with an entry threshold depending on its distance from the maximum-stiffness point.

Corollary 8.1 (A compact subarc basin)**.** *Fix $0 < d \le \pi$. There is a finite $K_0(d)$ such that, for every original $K \ge K_0(d)$, the entry conditions*

$$|\varphi_K + \pi/2| \le \pi - d, \qquad |n_K| + |u_K| \le h_K^2, \qquad |w_K| \le h_K^3 \tag{142}$$

imply annular capture, convergence of both actual sequences to $(0,-1)$, and finite length of both actual paths. The lift remains in

$$|\varphi_k + \pi/2| < \pi - d/2. \tag{143}$$

The normal, filtered-velocity and angle-remainder estimates of Theorem 2 retain constants independent of K and the starting angle. Theorem 3 holds for each fixed admitted entry, with no uniformity asserted as K varies or $d \downarrow 0$.

In particular, every zero-velocity unit-circle start other than $(0,1)$ selects $(0,-1)$ when its original starting time is sufficiently late. The threshold is uniform on every fixed compact subarc avoiding $(0,1)$.

Proof. Sections 8.1–8.4 did not use $\varphi_K \in [-\pi, 0]$. They therefore give the same actual angle equation and total error budget for (142). Choose any integer $K_0(d)$ strictly greater than

$$\max\{h_*^{-3},\ L,\ (8C_R/d)^3\}. \tag{144}$$

Then $2C_R h_K < d/4$. On $|\theta| < \pi$ the drift is restoring and the unperturbed step cannot cross zero. At every finite prefix before a possible exit from (143), (131) yields

$$|\theta_k| \le \pi - d + \sum_{j=K}^{k-1} |r_j| < \pi - 3d/4 < \pi - d/2.$$

This closes the strict barrier. Summation gives $\sum |f(\varphi_k)|/k < \infty$ and finite angular variation. The angular limit lies strictly between the adjacent maximum-stiffness angles. Harmonic divergence and the unique drift zero in that interval force $\theta_k \to 0$. Sections 8.6 and 8.7 now apply to prove the remaining claims.

For a unit-circle point other than $(0,1)$, choose the unique lift with $-\pi < \varphi_K + \pi/2 < \pi$. Its distance from the two endpoints is positive, so some $d > 0$ meets (142). The endpoint case $d = \pi$ is also covered: the initial angle is the minimum-stiffness angle, with the permitted small history and normal perturbations. ∎

This is a compact-subarc result with a distance-dependent late-start threshold. It does not classify the full phase-space basin or perturbations started at the excluded maximum angle.

9 Different schedules from the identical history

The uniform circle theorem gives a precise comparison with the reciprocal-clock prepared-circle result in [6]. Hold fixed the objective (79), $b = 1/4$, $\gamma = 3/2$, the original entry integer K, and the identical prepared history

$$x_{K-1} = x_K = \left(1 - \frac{1}{KH(0)}\right)(1,0) = \left(1 - \frac{1}{2K}\right)(1,0). \tag{145}$$

Run the same literal two-line recursion twice, with respectively

$$\begin{aligned} \text{old clock:} \quad & h_k = k^{-1/2}, \quad \varepsilon_k = 1/k, \\ \text{critical clock:} \quad & h_k = k^{-1/3}, \quad \varepsilon_k = k^{-2/3}. \end{aligned} \tag{146}$$

Changing this schedule changes both regularization h_k^2 and damping γh_k. The gradient-history coefficient, objective and history pair are unchanged.

For the reciprocal clock, the prepared-circle result in [6, Theorem 6, Section 4.8] applies to precisely (145) and proves

$$|\varphi_*^{\mathrm{old}}(K)| \le C_* K^{-1/2}, \qquad p_*^{\mathrm{old}}(K) = e(\varphi_*^{\mathrm{old}}(K)), \tag{147}$$

where C_* is independent of the sufficiently late original K. After a common threshold is enlarged, $|\varphi_*^{\mathrm{old}}(K)| < \pi/4$. In particular, $p_*^{\mathrm{old}}(K) \to (1,0)$ as $K \to \infty$ through these independent prepared histories. This is the exact old theorem used for the comparison; it is not an old-clock statement about the unprepared history $(1,0),(1,0)$.

For the critical clock, the same history has

$$\varphi_K = 0, \qquad n_K = -1/(2K), \qquad u_K = w_K = 0.$$

For every $K \ge 1$, $1/(2K) \le K^{-2/3} = h_K^2$, so it meets (84). Theorem 2 therefore gives, for every K above the two theorems' common finite threshold,

$$p_*^{\mathrm{critical}}(K) = (0,-1). \tag{148}$$

This does not require Corollary 8.1. The actual limiting points for the two schedules are different, despite the identical starting history, and

$$|p_*^{\mathrm{old}}(K) - p_*^{\mathrm{critical}}(K)| \longrightarrow \sqrt{2} \qquad (K \to \infty). \tag{149}$$

The old estimate is imported only for this comparison. It plays no role in the new-clock capture, selection or finite-length proof. The source archive includes the unchanged reciprocal-clock companion PDF as the comparison dependency. The comparison establishes different limiting points, not a claim that the new schedule is a faster optimization method.

9.1 The sharp upper normal endpoint

The normal range used by the general theorem has a sharp upper endpoint for a universal bounded-capture statement. The following is the critical-clock specialization of the endpoint theorem proved in the reciprocal-clock companion; its full conformal-metric proof is not repeated.

Corollary 9.1 (Critical-clock endpoint consequence)**.** *For $F(x) = 4x^2$ on $\mathbb{R}$ and $M = \{0\}$, the normal Hessian is 8. Under (4) with (3), every nonzero history $\mathsf{Z}_K = (x_K, x_{K-1})$ at a sufficiently late original K satisfies*

$$\frac{|\mathsf{Z}_k|^2}{\displaystyle\prod_{j=K}^{k-1}(1+\tfrac{3}{2}h_j)(1+\tfrac{1}{4}h_j^2)} \longrightarrow C(\mathsf{Z}_K, K) \in (0,\infty), \tag{150}$$

and

$$\log|\mathsf{Z}_k| = \tfrac{9}{8}k^{2/3} - \tfrac{21}{16}k^{1/3} + \tfrac{9}{16}\log k + O(1), \qquad \frac{\log|\mathsf{Z}_k|}{\sum_{j=K}^{k-1} h_j} \to \tfrac{3}{4}. \tag{151}$$

Thus the consecutive-pair norm diverges and the scalar sequence is unbounded. Arbitrarily small independent entries obeying the scaled normal-entry bounds can have this behavior. No conclusion that every individual $|x_k|$ tends to infinity is asserted.

Proof. Apply [6, Theorem 9] to the critical schedule. The example has a compact zero-dimensional minimum manifold and is globally defined. For the entry assertion choose $x_{K-1} = x_K = \delta$ with $0 < |\delta| \le \theta h_K$; then $n_K = \delta$, $u_K = v_K = 0$. The cited positive product limit gives the claimed unboundedness for these entries. ∎

9.2 Scope and reproducible algebra

At $\varphi = \pi/2$ in the circle model, $H' = 0$ and the positive vertical radial axis is invariant under both literal algorithm lines. The uniform radial capture proof applies at this angle as well. For sufficiently late equal unit histories at $(0, 1)$, or radial histories satisfying the same small caps, it keeps the orbit on that positive ray and gives convergence to $(0, 1)$. This is an exact exception to minimum-stiffness selection, not an assertion about all perturbations at the excluded angle.

All circle points remain minimizers of the objective. The compact-subarc theorem does not classify the full phase-space basin, and its threshold is not uniform as the initial angle approaches maximum stiffness. The fixed-entry scaled limit in Theorem 3 may vanish. The identical-history comparison changes damping and regularization together; it is not a speed comparison. No arbitrary-early capture, global least-norm selection, or schedule family beyond the two displayed clocks is asserted.

The supplementary scripts check the collision-safe metric, the normal target, finite raw-map coefficients, clock identities, the factor $b/\gamma = 1/6$, the deterministic KL augmentation algebra and the circle derivative $-1/54$. They are exact finite checks, not trajectory simulations or substitutes for the infinite-time proofs. The main critical-clock theorem and the circle selection proof need no reciprocal capture assumption. The only external mathematical dependencies within the paired manuscripts are the comparison and endpoint results explicitly cited in this section.

References

[1] H. Attouch, A. Balhag, Z. Chbani, and H. Riahi. Accelerated gradient methods combining Tikhonov regularization with geometric damping driven by the Hessian. *Applied Mathematics & Optimization* **88**, article 29 (2023). doi:10.1007/s00245-023-09997-x. Final published version, Section 4.2, p. 45, boxed IGATH-2.

[2] P.-A. Absil, R. Mahony, and B. Andrews. Convergence of the iterates of descent methods for analytic cost functions. *SIAM Journal on Optimization* **16** (2005), 531–547. doi:10.1137/040605266. Lemma 2.1, p. 533. Author-hosted published text.

[3] C. T. Woodward. Localization for the norm-square of the moment map and the two-dimensional Yang–Mills integral. *Journal of Symplectic Geometry* **3** (2005), 17–54. Appendix A, Theorem A.1 and the following paragraph, p. 48. Publisher full text.

[4] J. Bolte, S. Sabach, and M. Teboulle. Proximal alternating linearized minimization for nonconvex and nonsmooth problems. *Mathematical Programming* **146** (2014), 459–494. doi:10.1007/s10107-013-0701-9. Uniformized inequality in the author manuscript, Lemma 3.6, pp. 20–21.

[5] K. Kurdyka. On gradients of functions definable in o-minimal structures. *Annales de l'Institut Fourier* **48** (1998), 769–783. doi:10.5802/aif.1638. Theorem 1, p. 775, and definable monotonicity and differentiation facts on pp. 772–774. Primary full text.

[6] *Sharp normal stability and finite actual travel for reciprocal-clock IGATH-2.* Companion manuscript (2026). Theorem 9 supplies the endpoint result for both clocks; Theorem 6 and Section 4.8 supply the reciprocal prepared-circle comparison. The unchanged PDF is included with the sources.